Global Collaboratory on the History of Labour Relations, 1500 - 2000

Dataset: Russia

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General Remarks

This data-set on labour relations in Russia (for exact description of territorial coverage see section below) has been compiled within the framework of the Global Collaboratory for the History of Labour Relations, a project of the International Institute of Social History (Amsterdam, The Netherlands) and the Institut für Wirtschafts- und Sozialgeschichte (Vienna, Austria), with financial support by the Gerda Henkel Stiftung (Düsseldorf) and the Netherlands Organisation for Scientific Research (NWO). Using a universally applicable taxonomy elaborated within the framework of this project (cf. Appendix I) it classifies the population of Russia for five historical cross-sections in terms of labour relations. Starting from the broadest possible definition of work as "any human effort adding use value to goods and services"[1], from hunting-gathering and independent production to household work and slavery, the term labour relation is used as a common denominator for the wide variety of hierarchical, non-hierarchical, familial, forced, voluntary, monetary, tributary and other social relations under which this work is performed.

Crucial in this approach and in the taxonomy used, is that there is a labour relation for each and every individual, varying from "no labour relation" for people classified as non-

[1] Tilly, Chris and Charles Tilly, *Work under capitalism* (Boulder, Colo, 1998), p. 23

working due to age (too young or too old), physical disability, or the absence of any need to work (affluence) to a whole range of different possibilities for those employed, mobilised, assisting family members or otherwise engaged in the addition of use value to goods and services through the application of work effort. A second crucial principle underlying the taxonomy and the database produced is that it deliberately does not qualify the extent of work performed - full-time and part-time work are treated as equals, and the same applies to the systematic running of a household and repeated occasional help to in running the household. As a result of these two principals the sum of the different categories of labour relations in the database is therefore always equal to the total population.

The database has been compiled on the basis of population censuses and proto-censuses and consists of five individual data-sets for the following cross-sections: "1500" [1495], "1650" [1678], "1800" [1795], "1900" [1897], "2000" [2002]. The years indicated in square brackets are the exact years to which the sources used for their compilation relate. Detailed explanations on the sources used and the classification procedures applied are listed below for each historical cross-section.

The database is the work of three authors. Dmitry Khitrov is the author of the databases for the cross-sections 1500, 1650, and 1800 and Gijs Kessler and Olav Hofland have compiled the databases for the cross-sections 1900 and 2000. For more information on the authors and contact information users are referred to Appendix V, as well as to the web-site of the Global Collaboratory for the History of Labour Relations.[2]

Use of the database is subject to the conditions of the Licence Agreement included in Appendix VI.

Territorial Coverage

One of the principal challenges in charting the historical occurrence of labour relations in Russia is the tremendous expansion of the territory governed by the Russian state over the five centuries covered by the database. This is a challenge not only in terms of the availability of sources, but also at a more principal level. To achieve comparability over time it is desirable that the territorial coverage of the data-sets for the respective cross-sections do not differ too significantly. Because of the changes in territory over time, however, this inevitably means that for certain cross-sections the territorial coverage of the database offers a less than perfect match to that of the Russian state at that particular moment in time. Depending on the cross-section the territorial coverage of the database can be either smaller or larger than that of the Russian state. Of these two possibilities, the latter is the less preferable, because it would involve drawing in sources on these outlying areas, which might be non-existent, or compiled within statistical and bureaucratic traditions very different from those of the Russian state, and might therefore be very difficult to meaningfully integrate into the database.

Taking into account the different considerations, the authors have opted for a territorial coverage which roughly coincides with the 1800 boundaries of the Russian state, incorporating Siberia and the Far East (including territories acquired in the course of the 19th century), as well as most of modern-day Belarus, Ukraine and the Baltic States, but excluding Central Asia and the Caucasus.

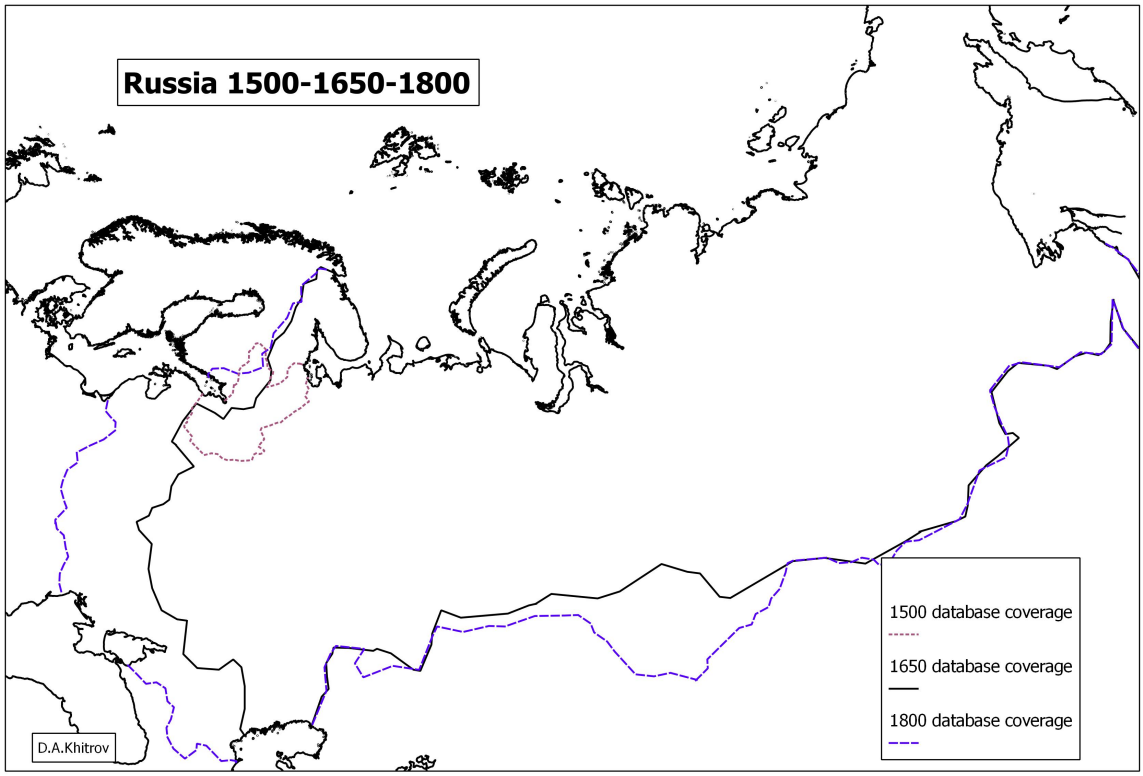
For the 1900 cross-section a similar territorial coverage can be achieved by excluding the data for Central Asia, Poland, Finland and the Caucasus from the aggregates for the Russian Empire as a whole, leaving only the subregions referred to at the time as European Russia and Siberia. For the cross-section 2000 this can be achieved by the aggregation of census data

[2] History of Labour Relations, 1500-2000 [https://iisg.amsterdam/en/data/data-websites/history-of-labour-relations-1500-2000, as retrieved on 20 September 2019].

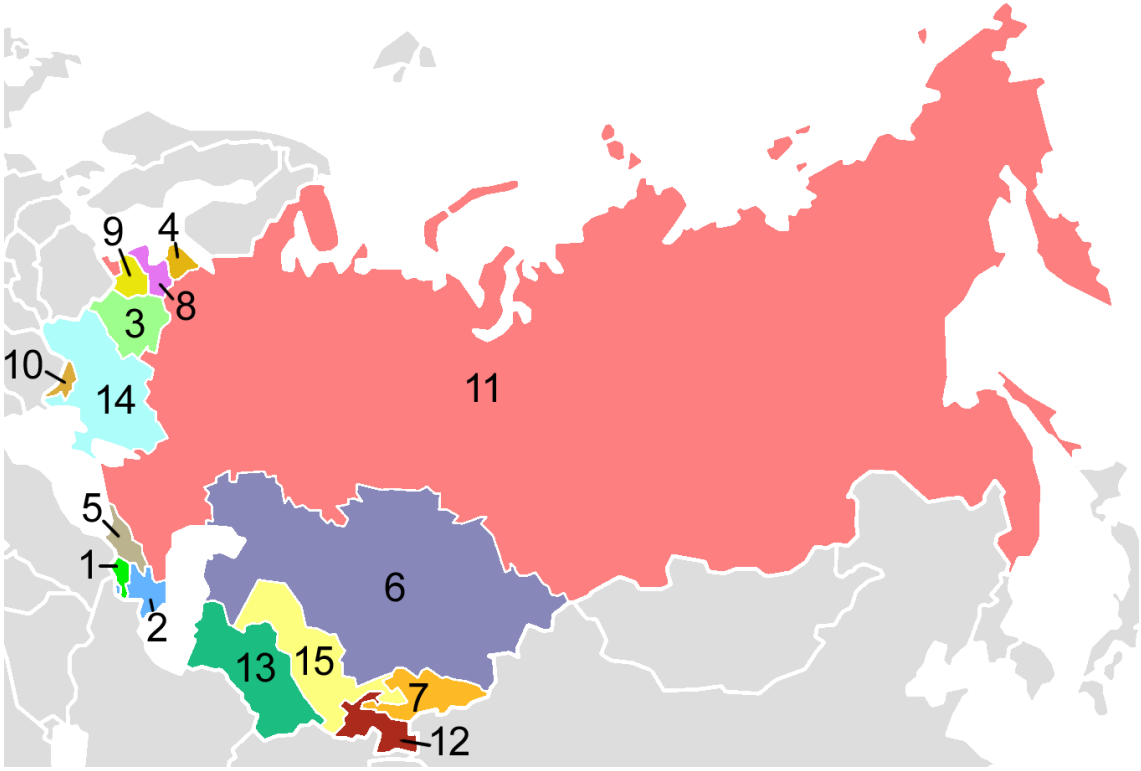
for the Russian Federation, Belarus, Ukraine, Moldova, and the Baltic States - all gathered within similar statistical traditions because of their recent common past within one and the same statehood and because of the increasing global standardization of census categories and procedures.

Backward comparability to the 1650 cross section is marred by the fact that the data for the cross-section 1650 do not cover the Ukrainian lands and the steppe north and east of the Black Sea which were added to the empire only later. Having said this, nonetheless the match between the territory covered by the 1650 and the 1800 data is close enough to warrant comparisons at a more general level of aggregation like the overall subdivision by labour relations. The cross-section 1500 remains the major outlier. Only data for a part of the Novgorod lands have withstood time, but these can be compared to data for the same lands in the 1650 cross-section, thus obtaining a second tier of comparability over time.

Map 1 - Territory covered by the databases for 1500, 1650 and 1800.



Map 2 - Russian Federation and other countries of the former Soviet Union, 2000[3]



1. Armenia
2. Azerbaijan
3. Belarus
4. Estonia
5. Georgia
6. Kazakhstan
7. Kyrgyzstan
8. Latvia
9. Lithuania
10. Moldova
11. Russian Federation
12. Tajikistan
13. Turkmenistan
14. Ukraine
15. Uzbekistan

[3] CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=109604

Comparability of the data over-time is summarised in the table below:

	1500	1650	1800	1900	2000[4]
Upper level of aggregation	Not available	Not available	Russian Empire	Russian Empire excl. Poland, Finland, Caucasus, Central Asia	Russian Federation + Ukraine + Moldova + Belarus + Baltic States
Medium level of aggregation	Not available	Muscovy state	Not available	Not available	Russian Federation
Lowest level of aggregation	Novgorod lands	Novgorod lands	Not applicable	Not applicable	Not applicable

Methodology

The data-set comes as separate MS Excel spreadsheets for each cross-section.[5] The database is compiled using the standard methodology described in the documentation for the Global Collaboratory on the History of Labour Relations.[6] Each record of the data-set stands for one or more persons identifiable by one and the same labour relation AND sharing a number of other characteristics, such as sex, age, profession, regional provenance, social status, or ethnicity. Each individual figures in the database only once, therefore the database in itself does not contain any subtotals. No single record, for example, exists in the database for the total number of women working in hired labour, regardless of profession. Instead the database contains separate records for each group of women sharing the same profession and the same labour relation.

The essence of the procedures followed in compiling the database is to first isolate the non-working (Labour Relation 1) from the working people and to subsequently divide the working population over the other labour relation on the basis of information on their social status, profession and other characteristics.

The separation of the working from the non-working combines data on the disabled and those otherwise unfit for work with data on the age-structure of the population. Unless indicated otherwise by qualitative and/or quantitative evidence it is assumed that in most societies people of 75 years or older, as well as people under seven years of age are not working, and are not expected to do so. The age-delimiters for inclusion into the category non-working vary somewhat between the cross-sections, depending on differences in longevity, schooling and expectations towards children and adolescents:

[4] Data for Ukraine, Moldova, Belarus, and the Baltic states are not included in this data-set, but compiled as separate data-sets. As of April 2019 a data-set for Belarus is available, others will follow in the future.
[5] Kessler, Gijs; Khitrov, Dimitry; Hofland, Olav, 2016, "Russia 1500,1650,1800,1900,2000 [Global Collaboratory on the History of Labour Relations 1500-2000 Dataset]", http://hdl.handle.net/10622/PLGUCD, IISH Dataverse, V2
[6] Hofmeester, Karin; Lucassen, Jan; Lucassen, Leo; Stapel, Rombert; Zijdeman, Richard, 2016, "The Global Collaboratory on the History of Labour Relations, 1500-2000: Background, Set-Up, Taxonomy, and Applications", http://hdl.handle.net/10622/4OGRAD, IISH Dataverse, V1

	1500	1650	1800	1900	2000
Lower age-delimiter	≤ 6	≤ 6	≤ 6	≤ 6	≤ 14
Upper age-delimiter	≥ 60	≥ 60	≥ 60	≥ 75	≥ 75

The database has been set up in such a way, however, to allow for the calculation of Labour Relation 1 using alternative age-delimiters, for example by integrally assigning all of the population aged 7-10 to the non-working category, if it is considered that only at eleven years of age children would start to pursue a trade or contribute to the household budget.[7]

As far as the working population is concerned, the database also provides information, wherever available, on multiple labour relations. People combining two or more jobs or economic activities can be simultaneously engaged in several different labour relations. Examples would be a peasant smallholder who works as a hired labourer during part of the year, a serf fulfilling his or her labour obligations to the lord during only part of the year while providing in his or her own existence during the remainder of the year, or indeed a nominally free population having to perform certain labour obligations to the local community or the state during part of the year.

To accommodate for the existence of such multiple labour relations per individual, the database offers the option to assign up to three labour relations for each entry. The relative importance of each of these labour relations for persons concerned can be indicated by providing information on the share of total working they accounted for. Because such information is extremely difficult to get by, in particular for the earlier cross-sections, the Russian data-sets offer a slightly different solution, not assigning percentages of time spent to the respective labour relations, but simply providing a ranking of their importance by putting the principal labour relation in the Labour_rel 01 field, the second most importance in the Labour_rel 02 field, and the least important in the Labour_rel 03 field.

The relative importance of the respective labour relations is assessed on the basis of additional qualitative and quantitative evidence on one or more of the following criteria: contribution to total income or total household income, share of time allocated to the particular activity, or the importance of the activity in terms of social and legal identity. Ideally, the ranking weighs all three elements, so as not to put in first place an activity which brings in substantial income but is really a side-activity in all other respects, or an activity which is of overriding importance for legal or social identity, like serfdom, but can actually affect labour relations of the individuals concerned during only a minor part of the year.

[7] Please, note that in the 1800 database a slightly different age-breakdown is adhered to: 0-6, 7-11, 12-14, 15-60, 61-99, as compared to 0-6, 7-10, 11-14, 15-60, 61-99 in the other data-sets.

Cross-sections

The following sections of this methodological paper provide comprehensive and full details on the sources used and the classification, coding, estimation and calculation procedures adhered to in compiling the database. For the convenience of the user this information is listed separately for each cross-section.

1500

[NB: the 1500 data-set was compiled using a previous version of the taxonomy, cf. Appendix II]

In 1500, the Grand Princes of Moscow were taking the last steps in uniting all the Russian principalities under their power (with some exceptions: Pskov and Ryazan still remained formally independent, and Smolensk and Chernigov were part of the Grand Principality of Lithuania). The territory of Muscovy was comparatively small, less then 1/5 of Russia's territory by 1650, but it was on the threshold of a period of rapid growth. The economic, demographic, and military potential of Russia in this period therefore presents an interesting historical problem, which has been a subject of discussion among several generations of historians.

Unfortunately, the sources at our disposal provide little hope of having these questions finally solved. Although a unified system of taxation was introduced throughout the possessions of Ivan III in the late XVth century and all tax-payers were registered, the records of this taxpayers' census [*писцовые книги*] vanished in the fires which periodically devastated the medieval Russian archives.

The sole exception are the materials concerning the region of Novgorod, where the census was held in 1495-1504. These materials have been integrally published and they have been investigated in several dedicated studies.[8] Although not all the social groups were recorded, we can make a well-founded estimate of the total population, as well as of the major social groups. Moreover, the materials provide some information on occupations and labour relations.

These figures cannot be extrapolated to other regions of the country, however. Novgorod was a very specific part of Russia at that time. Conquered by Ivan III in 1478, it was still not entirely integrated into the political and administrative structure of Muscovy, and there is some controversy whether it was more or less populated and economically developed than the regions of the Russian Heartland. The total loss of materials concerning all the territories except Novgorod rules out any source-based assumptions on this question.

Our first task was to assure comparability with the cross-section 1650. The former territory of the Novgorod republic was one of the biggest administrative provinces [*уезд*] of the Muscovy state. It was subdivided into five districts, called 'piatinas' ('one-fifth part'). The territory of the 1495 Novgorod *uezd* roughly coincides with the territory of two *uezds* in 1678 (the benchmark-year for our next cross-section): Novgorod and Staraya Rusa.[9] There are two big mismatches, however. In the first place, the Baltic seaboard, with the fortresses of Yama, Ivangorod, Koporie, Oreshek, and Korela, was assigned to Sweden according to the terms of the Stolbovskij peace treaty of 1617. According to XVIth century data these lands had no significant rural population. Therefore, we have excluded the population of those 5 forts from the database to ensure comparability with the data for 1678. Secondly, we also have no data for the city of Novgorod itself before 1580s. Any extrapolations are too risky, as they would have to bridge the catastrophe of 1570, when this medieval megapolis was

[8] Cf.: Stepanova L.G., *Novgorodskoe krest'ianstvo na rubezhe XV-XVI stoletii.* (Moscow, 2004) (bibliography).

[9] Another part of the 1495 Novgorod *uezd,* the territory of the *Zaonezhskie pogosty* had become part of the uezd of Olonets by 1678, but as its census books of 1495-1504 are lost, we cannot include it into the database for 1500. Cf.: Nevolin, K.V., *O piatinakh i pogostiakh Novgorodskikh.* (St. Petersburg, 1853) - Map; Vodarskii Ia.E. *Naselenie Rossii v kontse XVII - nachale XVIII veka* (Moscow, 1977), pp. 252-253.

captured and devastated by Ivan Grozny's *oprichniks*. Consequently, we have to limit ourselves to the rural territories, and mark all database records for 1500 as Non-Urbanized.

The census books survived for only 3 of the 5 piatinas. This is a serious problem, because for the cross-section 1650 we only have the data for the Novgorod and Staraya Russa uezds taken together. For the other two we can use existing estimates from historiography, in particular from the fundamental 'Agrarian History of Russian North-West'[10]. They are based on the materials of the mid-XVIth century and on a sophisticated system of extrapolation of population density and household size figures from the nearby territories. Consequently, the picture of occupational structure is much more generalized for the territories where we use extrapolations, instead of direct sources.

The sources give us the number of households and families, but no figures on population size.[11] In order to obtain a figure for total population size, we use the advanced calculations of 'Agrarian History', which take into account both of the indicators just mentioned.[12] Average household size was calculated using the assumption that there would be 5 men and women if there was only one family within the household, 7,5 if there were two, 10 if there were three, etc.

The sources give us five types of households.

1. The households of the feudal lords. As a rule, the feudals in medieval Russia were obliged to perform military service, therefore we have marked Tributary labour [LabRel 8] as Lab_Rel 01 for all the heads of these households.

A more complex issue is whether they obtained their means of subsistence just by collecting rent from their peasants and by using the corvee labour of their slaves, or also by working their land themselves. Two broad categories existed: so-called *svoezemtsy*, whose land-titles dated to the period of Novgorod independence[13], and *pomeshchiks*, who had obtained their manors after 1478. For the territory covered by the sources they accounted for 53 % and 47 % of the total number of feudal lords respectively[14], which ratio we have extrapolated to our total figures. According to historiography, *svoezemtsy* usually worked their lands themselves (although some of them also extracted rental payments from their peasants), so they are marked as Labour Relation 4 - Household Leading and Kin producers, *Pomeshchiks*, to the contrary, did not work their own lands, but their military service was much more intensive. One question is, did all adult men serve in the army, or only part of them? The number of people which used to be sent to the army from each manor depended on its size. If we take the earliest recorded norms on this issue, those of the mid-XVIth century, and relate this to average manor size, this would result in an average of two men per manor performing military service would result[15], a figure which roughly equals the average number of men of working age per manor - in other words all men older than 15 and younger than 60 served in the military. Consequently, they all were classified as Labour Relation 8.

2. The households of slaves. Slaves were corvee labourers, as a rule, but they were also running their own households.[16]

3. Peasants formed the absolute majority of the population. They paid rent to the feudal lords, but did not generally work on the corvee fields *(gospodskaia pashnia* or *barshchina),*

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[10] *Agrarnaia istoriia Severo-Zapada Rossii. Vtoraia polovina XV - nachalo XVI v.* (Leningrad, 1971) (Hereafter - AISZR).
[11] Ibid., pp. 17-20.
[12] Ibid., pp. 322.
[13] They were also obliged to perform military service. See: Bernadskii, V.N., *Novgorod i Novgorodskaia zemlia v XV v.* (Moscow-Leningrad, 1961), pp. 125-128; AISZR, pp. 341-342.
[14] AISZR. p. 339.
[15] According to the Code of Service (1558), 1 horseman per 50 hectares of ploughland. The average size of manor within this group *(pomeshchiks)* is 112 'korobias', which is nearly equal to 110 hectars (AISZR, p. 335).
[16] AISZR. pp. 354-357.

which at that time was supposed to be the work of slaves.[17]

4. The 'Non-plowing' people *(nepashennye liudi)* included all peasant households which did not have their own plough-land. These were mostly artisans, but also impoverished peasants, beggars, and families where there were currently no members able to do field work. We have singled out the artisans[18], and marked all others as "non-working and not expected to work" [Labour Relation 1]. This is of course an exaggeration, but likely not too far from reality, as they account for less than 2 % of the total.

Village artisans can be identified only in those parts of the territory for which the census books have survived. Assuming that their share among the total population was the same in the other parts of the territory, we have calculated their total number, but have not specified their trades.

5. Orthodox clergy, unlike the Catholic one, had families and combined field work with divine service.

Next, we need to assess the size of the working and the non-working population. Unfortunately, the nature of our sources does not allow us to make any distinct calculations concerning the demographic characteristics of the population. The earliest reliable data for the reconstruction of the demographic pyramid are the censuses of the early XVIIIth century We have several samples based on these sources, and a fundamental study by E.N. Baklanova on the Vologda region for 1717.[19] We depart from the assumption that before 7 years old a child was not able to work to the extent that the results of its efforts were significant to the family. According to Baklanova, children of this age made up 17,6 to 19,2 % of the total population, and we set this value at 18 %. We have also singled out the groups of 7-10 (when the children began to assist their parents within the household) and 11-14 years old (when children were considered 'half-workers' and participated in all major adult work, except military and civil service). Each of these groups is estimated as 9 % of the total. Traditionally, among Russian peasants 15 was the age when a young man or woman was considered to have become a worker (*rabotnik*) in his or her own right, meaning that their labour started to make a significant contribution to the total effort of the household; also, 15 was supposed to be the minimum 'normal' age for marriage, and for paying taxes, when the community divided the total tax burden among households.[20] The age at which people ended their labouring life was subject to much greater variation, and it is hard to draw a distinct line separating the working from the non-working in this respect. Average life expectancy was very low – according to the same sources only 4,5 % of the population was older than 60. Considering this, it would not be a great exaggeration to classify all of them as non-working.

Finally, we have absolutely no data concerning the female population and nuptiality. Usually, researchers use the assumption, drawing on the first available evidence, which dates to the XVIIIth century, that the number of women among the population is approximately equal to the number of men, while marriage rates among the adult population tend to the maximum (due to early marriages, the absence of divorce, and a high probability of second marriage after the death of one of the spouses). There is no contemporary direct or indirect evidence for these assumptions, though, and reality can have been very different for certain social groups or distant regions, but these deviations cannot have been of the order of magnitude to have a significant impact on calculations and estimates at the aggregate level.

[17] AISZR. pp.. 350-353.
[18] AISZR, p. 318.
[19] Baklanova, E.N., *Krest'ianskii dvor i obshchina na russkom Severe. Konets XVII – nachalo XVIII veka.* (Moscow, 1976), p. 21.
[20] For an in-depth treatment of the issue, cf. Baklanova 1976, p. 40-44.

1650

[NB: the 1650 data-set was compiled using a previous version of the taxonomy, cf. Appendix II]

The basic data for any demographic or social research concerning mid-seventeenth century Russia can be obtained from the materials of the fiscal census of 1678/79 - the so-called 'perepisnye knigi'.[21]

The territory covered by the census was that of the Russian state of 1678, excluding the Left-bank Ukraine (*Getmanstvo*). The census was held in European Russia only, but similar Siberian sources exist for the 1680s and early 1690s. This corresponds well with the approach we use.

The materials of the 1678 census contain information on the main social groups obliged to pay taxes, i.e. peasants and tradespeople (*posadskie liudi*) as well as slaves and parish clergy. The census does not cover the nobility, monks, and several minor groups obliged to perform military service instead of paying taxes. Also, the numerous peasants belonging to the Court ('*dvortsovye*') were not recorded. Fortunately, all these data can be obtained from other contemporary sources. For the *'dvortsovye'* peasants, a similar census was held in the late 1670s and early 1680s. The numbers of people belonging to the social groups not enumerated in the census were calculated using the materials of the Razriad (Military Department), Patriarch's Chancellery, and other ministries which oversaw them.[22] Summarising, it is possible to obtain source-based data covering the entire population of the country.

At the same time, we face two serious problems in working with this material.

First, there was no centralized system of population registration nor a centralized budget in the XVIIth century Russia, and therefore the results of the census were never aggregated at the national level. For each of over 200 uezds 'a census book [*переписная книга*]' was compiled, but by now, only part of these books survived, so we cannot recalculate the total population directly from them. Many different calculations were carried out over the long period when these books were used as a taxation cadaster (until 1724), but none of them covers the whole territory of Russia and all social strata.

Secondly, a significant part of population avoided the census altogether in order to avoid taxation. This is quite common for censuses of the early Modern era in general[23], but it presents the problem of comparability with later cross-sections when censuses became far more accurate.

For these reasons the materials of the census cannot be used directly. Instead, we rely on the seminal work of Ya.E. Vodarskii, who traced missing figures and revised, recalculated and combined the various data in "The population of Russia in the late XVIIth – beginning of the XVIIIth century", published in 1977. Working on an atlas of historical demography, he faced exactly the same problems we do. Combining the different figures used by numerous civil departments to determine the number of tax-payers, the 'general tables' compiled during the reign of Peter the Great, and the figures from the remaining census books of 1678, he painstakingly reconstructed the overall results of the census. Subsequently, using the materials of the few governmental verifications of the accuracy of the data, he estimated the number of those who avoided census registration and taxation as 25%.

The only shortcoming of Vodarskii's work is that his calculations merely reveal the size

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[21] A census was also held in 1646-1647. It is much closer to the cross-section date, but it has vast lacunas, because standards of description were poor and the materials we have are incomplete. Also, its results were not aggregated at the time or in historiography, which would make it very-consuming to use. For these reasons we have opted to use the materials of the 1678 census for compiling the database for the cross-section 1650.

[22] Vodarskii, 1977.

[23] Cf. Vodarskii, Ia. E., "Perepisi naseleniia v XVII v. v Rossii i drugikh stranakh Evropy", in: *Feodal'naia Rossiia vo vsemirno-istoricheskom protsesse.* (Moscow, 1972).

of the main aggregate social strata, such as the nobility and the peasantry. But different labour relations could exist within each of those major groups. Therefore, if we simply rely on his figures, our picture of occupations and labour relations in Russian society will be too generalized.

One possible solution is to combine the estimates by Vodarskij with the data of partially aggregated primary sources. In 1708, when the census books of 1678 were still used for tax assessment, so-called 'general tables' *(generalnye tabeli)* were compiled, which aggregated the data of the census books and covered the central[24], western, north-western, and southern regions of Russia, together accounting for over 60 % of the total population. In these 'general tables' the social stratification is rendered in much more detail than in Vodarskij's research; these materials are published and can be used in our work.[25] Therefore, we use the data of the 'tables' for the part of the population they cover and rely on Vodarskii's figures for an estimate of the remaining part of the population. One of the consequences of this approach is that the occupational structure of this remaining part, like the population in the border regions of Russia as well as such social groups as 'dvortsovye' and 'chernososhnye' (state serfs), can be captured in much less detail than that of the population covered by the 'general tables'.

Summarising, the quality of the records in the database is homogenous, as they are derived from one and the same primary materials. But the level of detail varies among the different parts of the country, as well as among social groups.

The 1678 census focused on households, because these were the objects of taxation. They include also the names and number of men, particularly those bound to the soil, because the census books also recorded relations of personal dependence. Again, we assume the number of women to have been equal to the number of men. To separate the different age groups and to identify the work they do we use the same approach as for the cross-section 1500, based on the work of E.N. Baklanova.

The assessment of the rural/urban ratio among the population also presents some difficulty. Due to low population density and poorly developed internal trade, only a few major political and economic centers exceeded a population size of 5000, which is the benchmark value used for classifying a population as urban in the taxonomy of the Collaboratory.[26] The major part of the urban population belonged to the social group of tradespeople ('posadskie'); according to our calculations, average household size was 2.18 men, which would equal 4.36 men and women. Following this logic, if there were 900 households of tradespeople in a city, this would equal 4000 city dwellers. Assuming then, that in such an urban centre the non-taxpaying population would amount to a further 1000, we classify population centres with over 900 households in the *posad* as urbanized. Within the territories covered by the 'Tables' this applies to Nizhnij Novgorod, Yaroslavl, Kostroma, Vologda, Velikij Novgorod, Pskov, and their population is marked as 'Urbanized'. Unfortunately, another four significant urban centers existed outside the territory covered in the 'tables' – Arkhangelsk, Vyatka, Ustiug, and, alas, Moscow. We have no option to estimate

[24] With one significant loss – we have no tables for the city of Moscow.

[25] *Perepisi naseleniia Rossii: Itogovye materialy podvornykh perepisei i revizii naseleniia Rossii (1646-1858)*, Vyp. 1 (Moscow, 1972).

[26] The difference between a town (*gorod*) and a village (*selo, derevnia*) was clear to contemporaries, but has little to do with modern concepts and can therefore not be used in the database. Decisive were the administrative and military functions of a settlement, rather than its size or economic characteristics According to this logic, the center of an administrative district (*uezd*), or a significant border fort was always considered a town, whereas some significant trading centers such as Rogachevo or Pavlovo never obtained this status before the beginning of the industrial era. In the central regions, a community of merchants and craftsmen (*posad*) was an important attribute of a town, but not for example in the border regions or in Siberia. But in Siberia no settlement could be called a town if it did not have fortifications and a permanent garrison, whereas this would have been the exception rather than the rule in the cities of the European North.

their population, and it would be too risky to single it out-from the general estimates of Vodarskii. Consequently, levels of urbanization have been underestimated in the database..

The Collaboratory relies on the Historical International Standard Classification of Occupations (HISCO) for coding data on occupation, but our sources describe types of economic activity in a way very different from the approach realized in HISCO. Russian society in the mid-seventeenth century was divided into a number of social ranks (*chiny*), and each of them linked to the provision of a particular 'service'. The census identifies the rank according to the type of this 'service'. For example, the 'yamschiks' are not simply 'transport workers', which would be a literal translation of the word, but those who perform 'transport duties' for the state. The problem is, however, that they could easily derive their means of existence from agriculture, rather than from work in transport, and this is not an exception, but rather the rule.

We have accommodated for this in the database by identifying each social rank in the 'Position' field, while giving an interpretation of their occupations in the 'Branch' and 'Occupation' fields.

Further difficulties are encountered when interpreting labour relations.

Of the three major divisions offered by the taxonomy of labour relations used in the Collaboratory, two at least can be applied to the absolute majority of the population described in our sources.

First, everyone had his own household or lived in someone else's household. Rather than a type of work in the sense of the Collaboratory, this was a manner of living. The storage of stocks, animal husbandry and preparing food were activities present in any household, and in the absolute majority of them it was combined with field work. Even the nobles spent much effort running their households and ruling their manors, and the *Domostroy*, the mid-sixteenth century code of Orthodox ethics and one of the most popular and respected books at the time, calls running the household in the proper way not only the way to prosperity, but also a condition for eternal salvation. Among artisans there was a division between the 'plowing' *(pashennye*) and 'non-plowing' *(nepashennye)*, and the latter were numerous only among highly skilled professions like jewelers or clock-makers. But even within such groups, the women and children would work within the household as kin producers, running the kitchen gardens, raising livestock and small fowl, etc. Therefore, in the column Labour_rel_01 all the people living within the households (except the few 'non-plowing' artisans, administrators, etc) should be classified as reciprocal labourers. The number of leading household producers is assumed to have been equal to the number of households, and the number of household kin producers are calculated as the number of people minus the number of households. As a rule, households were run by a family couple, and the leading producer was the man; an exception has been made for the households marked as belonging to widows. The number of kin producer was calculated as the number of working people minus the number of leading producers.

Secondly, tributary labour was very widespread in Russian society. All subjects of the Tsar were obliged to perform services to the state (*sluzhba*) when needed. For some social groups such services were episodical (although sometimes very heavy), and our sources do not reveal them. Others carried them out as a permanent duty instead of or in addition to paying taxes, and were marked as 'Tributary Serfs". Serfs in the estates of the nobility, the bishops and the monasteries paid both quitrent *(obrok)* and worked in the corvee fields. Historiography holds it that the corvee was present in the absolute majority of estates by the second half of XVIIth century, even if the size of the burden could vary[27], and therefore all

[27] Cf. *Istoriia krest'ianstva SSSR s drevneishikh vremen do Velikoi Oktiabrskoi sotsialisticheskoi revoliutsii*. (Moscow, 1993), Vol. 3., p. 153; Tikhonov, Iu.A., *Pomeshchich'i krest'iane v Rossii* (Moscow, 1974), pp. 293-294; Gorskaia, N.A. *Monastyrskie krest'iane Tsentral'noi Rossii v XVII v.* (Moscow, 1977).

serfs have been classified as 'Tributary serfs'. Slaves consisted of two major subgroups. 'Dvorovye' were servants on the nobles' estates. Some of these lived in with their master's households, whereas others had their own households, but they all were supposed to be 'non-plowing'. 'Delovye', 'zadvornye', 'krepostnye' and some other groups were actually peasants, not servants, although they were legally bound to their master, rather than to the soil. They had their own households, worked on the corvee fields and were not, as a rule, used as servants. Consequently, they have also been classified as 'Tributary Serfs'.

The nobles were obliged to render military service from the age of 15 onwards, with no right to retire until they died or were removed from service as invalids. Attempting to avoid the service resulted in the confiscation of lands and criminal prosecution. Their position was not far from the position of conscripted soldiers, and therefore they are classified as indentured tributary labourers. Clerics are, again, marked as household producers and wage-earners working for non-market institutions.

The census provides no meaningful data concerning commodified labour, although it did exist. Hired labourers were not very numerous, however (30 000 is a maximum estimate) and did not form a stable group within society, which makes it difficult to single them out in the database.

1800

[NB: the 1800 data-set was compiled using a previous version of the taxonomy, cf. Appendix II]

In the XVIIIth and the 1st half of the XIXth century accurate population counts were of vital importance to the Russian state, because the poll-tax was the major item of revenue to the budget. To this end all tax-paying subjects of the empire were periodically registered in a kind of a census, the so-called *revisiia*. First introduced by Peter the Great in 1719, subsequent revisias were carried out in 1744, 1763, 1782, 1795, 1811,1815, 1833, 1850, and 1857.

The materials of the 5-th revisia are the most appropriate source for this part of the database. This is for several reasons. The date of this census (1795) is closer to the benchmark date than any other, and its materials are evaluated by historiography as more accurate and complete than those of the 4-th and the 6-th revisias. According to the estimates of V.M. Kabuzan, aggregation and calculation errors in these materials never exceeded 2 %, which can be considered an excellent result for the XVIII century. The revisia included all population of the empire except the nobility, the regular army, and foreigners (those who did not live in Russia permanently), whereas the quantity of the first two groups can be obtained from parallel sources, which has already been done in historiography.

The highest-level resulting documents of the revisia were the so-called *okladnye knigi* ('tax books'), the basic fiscal documents of the state, which were compiled annually. The tax book of 1808, generally acknowledged in historiography to be the most complete and accurate among those pertaining to the 5th revisia, has been integrally published and has been used for creating the database.[28] One important note is that population changes occurring between the revisias were not taken into account, which means that the book of 1808 still contains the data of 1795. Only those who arrived from abroad and became subjects of the empire within these 12 years were added, and they have been excluded from the database.[29]

The materials of the *okladnaya kniga* cover the whole territory of the empire, with the

[28] *Perepisi naseleniia Rossii* (1972), Vyp. 6.

[29] This is why the total figures in the database differ slightly from the figures by V.M. Kabuzan. Cf. Kabuzan, V.M., Narodonaselenie Rossii v XVIII - pervoi polovine XIX v. (po materialam revizii) (Moscow, 1963), pp. 159-160.

exception of the regions of Vilna and Grodno (in modern-day Lithuania and Belorussia); also, non-taxpaying citizens in Moscow and Slobodskaya Ukraine (in modern-day Eastern Ukraine, with the center in Kharkov) were omitted. The editors of the source publication retrieved those data from the lower-level primary documents of the revisia, and printed them in the same volume. A significant advantage of the book of 1808 is that the figures are bound to the administrative division of 1808, which is quite similar to that of 1897 and enhances comparability with the 1900 cross-section.

The primary record of the revisia, the so-called 'revizskaya skazka', included the following data: name, marital status, gender, age, social status, location. Unfortunately, not all these materials were aggregated. We have no general figures concerning age and marital status, so for this information we have to rely on estimates made by historians who worked with the primary data, as well as on some general assumptions concerning the upper and lower boundaries of the working age.. Moreover, only the male population was counted in the *okladnye knigi*, but we do know the total female population from the annual governors' reports.

Several steps were made to adapt the raw data to the format of the database.

1. The starting point for all further calculations is the total male population as listed in the *okladnaya kniga* of 1808 (and the additional data). These figures are broken down by region (*guberniia*) and social status.

When interpreting occupations and labour relations, the 'social status' column is critical. The system of social stratification in the late XVIIIth century was still quite sophisticated (although much less than that of 1678), and each social group is identified according to its position in society and to its obligations before the state. Some of them can be found in all regions (like state serfs), others exist only in several, or even one, guberniia (e.g. *mishari,* a social and ethnic group of non-Bashkir and non-Russian settlers in Orenburg guberniia). Referring to historiography and legislation (compiled in The Complete Collection of Laws of Russian Empire, 1830) this gives us the possibility to determine ethnicity, urban/rural residence, and sector of the economy (agriculture, trade, industry, or civil service) for each of these social groups, and to subsequently arrive at some assumptions concerning the corresponding labour relations.[30]

In some obvious cases some standardization was applied to the terminology of the sources to facilitate the use of the database. We have uniformly marked groups sharing a set of characteristics but historically having different names in different regions; for instance, in Ryazan 'the state serfs, called in the guberniia by their old name as *prezhnih sluzhb sluzhilye ludi, kazaki, streltsy, pushkari, razsylshiki, vorotniki, and pashennye soldaty'* were marked as 'state serfs'. In the same way, the numerous groups of peasants bound to the military service were unified as a 'service class' category *(voinskie obyvateli)*.

Next step is to add several groups missing in the *okladnaia kniga*.

The number of nobles was calculated by V.M. Kabuzan and S.M. Troitskij in their study[31] on the basis of the 'geographical descriptions' *(topograficheskie opisaniia)*, a series of geographical studies, compiled in the 1780s-1790s. The size of the field army and the garrison population were calculated by L.G. Beskrovnyi on the basis of the documents of the Voennaia Kollegiia (Ministry of Defense).[32] Here, there is a risk of double-counting, because some nobles served in the army; additional evidence[33] allows us to estimate their number at

[30] *Polnoe sobranie zakonov Rossiiskoi imperii.* Sobranie 1. Vols. 1-45 (St. Petersburg, 1830).
[31] Kabuzan, V.M., Troitskii, S.M., "Izmeneniia v chislennosti, udel'nom vese i razmeshchenii dvorianstva v Rossii v 1782 - 1858 gg"., Istoriia SSSR, 1971, Nº 4.
[32] Beskrovnyi, L.G., *Russkaia armiia i flot v XVIII veke.* (Moskva, 1958).
[33] Cf. Shepelev, L.E. *Chinovnyi mir Rossii XVIII - nachala XX vv.* (St. Petersburg, 2001), p. 113; Mironov, B.N. *Sotsial'naia istoriia Rossii* (St. Petersburg, 1999), Vol. 1, p. 134; Volkov, S.V., *Russkii ofitserskii korpus* (Moscow, 1993), pp. 269-270.

9600. The navy presented a more challenging task, because no reliable figures can be obtained from historiography. We have used the 'staff list' of 1798, although this could cause some overestimation, as these official staff positions were almost never filled for the full 100%. Finally, the indigenous population of the Russian Far East (Priamur'e and Primor'e) is added to provide territorial compatibility with the later cross-sections. This population was estimated as 38 000 in the late 17th century, significantly decreased in the wake of several invasions by Qing troops, and slowly increased again afterwards to reach 16000 in 1857.[34] For 1795 we have stuck to an estimate of 14000 (7000 men)..

Thus, as far as the male population is concerned, all information entered into the database is procured directly from the sources.

2. The next step is to separate those who work and those too old or too young to work. The age-delimiters used are the same as for the earlier benchmarks.

Unfortunately, the data on age from the primary documents of the revisiia were never aggregated, and, again, the only way to determine the size of the age groups is to extrapolate the data concerning individual manors, which have been gathered in historiography.

A number of such samples, representing different regions over the period from the 1760s to the 1790s, can be obtained in the works of Yu.A. Tikhonov[35] and M.F. Prokhorov[36].According to these authors, the percentage of the 0-15 years age-group varied from 34 to 47 %, with an average of 37.5%, and the percentage of the 60+ group from 3 to 6%, with an average of 4.5 %. This is very close to the data of E.N. Baklanova for 1717, used for the earlier benchmarks, and of Steven L. Hoch for the 1820s-1840s (45.65 % of the non-working population).[37]

Accordingly, we have estimated the relative sizes of the different groups as specified below:

0-6 – 18,75 %
7-11 – 11,72 %
12-14 – 7,03 %
15-60 – 58%
61+ - 4,5%.

3. The third step is to add the values on Urbanisation.

From the point of view of tax-collectors, the population of cities and towns consisted of two social groups – 'merchants' (*купцы*) and 'townspeople' (*мещане*). Historically, this has never been the case, though, and the population of towns and cities always included nobles, clerics, their servants, garrisons, as well as some peasants; also, since the late 18th century there was significant labour migration to the cities. There are no possibilities to calculate or estimate the numbers of people involved in these additional categories, and therefore the

[34] See: Dolgikh, B.O., *Rodovoi i plemennoi sostav narodov Sibiri v XVII veke.* (Moscow, 1960); Kabuzan, V.M. *Dal'nevostochnyi krai v XVII – nachale XX v. (1640-1917)* (Moscow, 1985), p. 46.
[35] Tikhonov, Iu.A., *Dvorianskaia usad'ba i krest'ianskii dvor v Rossii 17 i 18 vekov: sosushchestvovanie i protivostoianie* (Moscow, 2005), p. 329.
[36] Prokhorov, M.F., "Krest'iane obrochnykh votchin Rostovskogo uezda v seredine XVIII v. (po materialam podvornykh opisanii)", in: *Istoriia i kul'tura Rostovskoi zemli: Materialy konferentsii 1995 g.* (Rostov, 1996), p. 140-146; Prokhorov, M.F., "O genealogii krest'ianskikh semei v Rossii v XVII - pervoi polovine XIX (po materialam Pokrovskoi votchiny Naryshkinykh)", in*: Genealogicheskie issledovaniia* (Moscow, 1993), pp. 160-167; Prokhorov, M.F., "Krest'ianskaia sem'ia v krepostnoi derevne Podmoskov'ia v XVIII - pervoi polovine XIX v.", in: *Problemy istorii Rossii* (Ekaterinburg, 2005), Vol. 6, p. 233.
[37] Hoch, S.L., *Krepostnoe pravo i sotsial'nyi kontrol' v Rossii: Petrovskoe, selo Tambovskoi gubernii.* (Moscow, 1993), p. 65 and further.

actual level of urbanization will be underestimated in the database.

Having said this, the bulk of the urban population nonetheless consisted of merchants and townspeople. But here, again, we have to determine which part of them lived in settlements with more than 5000 inhabitants. It is impossible to do so on the basis of the aggregate data for each region (*guberniia*). Therefore, we have resorted to the use of the lower-level data for the individual *uezdy* (a *gubernia* typically consisted of 20-40 *uezdy*. As a rule, there was only one urban center in an *uezd*, and if the total number of merchants and townspeople exceeded 4000 (2000 men), it is highly probable that its total population exceeded 5000 (assuming that the other categories accounted for at least 1/5 of the settlement population).

Further investigation revealed that this method works for all the regions, excluding Ukraine and Belorussia. There, a significant Jewish population existed, which was invariably classified in the revisiia as merchants and townspeople, no matter if they lived in the towns and cities, or in the rural settlements, called 'mestechko'. Therefore, in these areas an extra criterion is required to determine whether an *uezd* center was a town with more than 5000 inhabitants. Using the method of trial and error we found this to be the case when there is more than a thousand non-Jewish urban population in a particular *uezd*. On these grounds we have excluded 32 *uezds*, while classifying the population of the remaining 129 population centres as "urban".

4. Next, we should determine which part of the household population should be classified as Household Leading Producers (Lab_Rel 04) and Household Kin Producers (Lab_rel 05).[38]

To calculate this, we need the number of households. Unfortunately, we have not found the aggregated data for 1795. We therefore rely on an estimate, which is informed by average household size. Average household size was calculated using the results of the census of 1710 and the materials of 1858, assuming that the pace and direction of change in the various regions was roughly the same.

Average household size[39]

	Year		
Region	1710	1858	1795
West	10.6	8.0	9.1
North-West	7.4	6.8	7.1
Center	7.4	6.8	7.1
North	6.8	7.4	7.1
East	6.6	8.2	7.5
South	7.8	10.2	9.2

Next, the number of households was calculated by dividing total population figures by average household size. Having thus obtained the number of households, the number of Leading Household Producers was taken as equal to it, and the number of Kin Producers was calculated as the total household population minus the number of households.

[38] The labour relation 'household servant' was very rare among the Russian peasantry. It could occur among artisans (and their apprentices), but our sources do not record such cases.

[39] Cf. *Istoriia krest'ianstva SSSR* (1993), p. 366; Struve, P. *Krepostnoe khoziaistvo. Issledovaniia po ekonomicheskoi istorii Rossii v XVIII i XIX v.* (St. Petersburg, 1913), pp. 224-227. The figures in the table refer to both men and women.

5. Serfs appear as a homogeneous social stratum in our sources. The landlord *(pomeshchik)* had the right to determine the type and amount of obligations of his 'subjects', and the state did not intervene in their relations, treating all the serfs as one estate. In reality, though, there was a great difference between peasants (*krestiane*), and servants on the nobles' estates (*dvorovye liudi*).[40] In 1861, the latter accounted for about 7 % of all serfs; we have extrapolated this figure into the late XVIIIth century in order to single out this category.

Serfs fell apart in two categories. On the one hand, there were those who paid a quitrent (*obrok*) in kind or cash to their legal owners and were then free to engage in whatever economic activity they saw fit. On the other hand, there were those who actually performed their feudal obligations on the land of their owner, the so-called 'barshinnye'.[41] It ought to be distinguished well between the two categories, because in the database the latter category ought to be classified as 'tributary serfs'.

The prevalence of those two forms of feudal exploitation differs much from region to region. For an estimate of this regional spread the calculations of V.I. Semevskij can be used, even though they are based on the earlier materials of the General Land Survey (*General'noe Mezhevanie*), which relate to the 1770s-1780s).[42] His calculations, covering more than half of all serfs, provide a percentage for both categories for each *guberniia*. Using these coefficients, we have split the lump-category 'Serfs' from the sources into 'Serfs (*barshinnye*)', and 'Serfs (*obrochnye*)'.

6. One important thing never recorded in the materials of the *revisiia* is the growth of wage labour. Being in the first place a fiscal census, as well as an instrument of consolidating the social division into estates, the revisiia did not even attempt to record people's means of existence. In the majority of cases, this can be reconstructed according to the major occupation of each social group. But wage labour was an economic pursuit for only minor parts of the major estates, and existed throughout - in cities and towns, and in small village industry. Moreover, the revisiia-also ignored seasonal labour migration (*otkhod*), quite widespread in the late XVIIIth century.[43] Historiography does present us with an abundance of estimates on this issue, however.

N.L.Rubinshtein summarized the numerous investigations into the historical emergence of a labour market in Russia which were conducted in the 1930s and 1940s, and arrived at an estimate of 220,000 for the total number of wage-labourers in the late XVIIIth century.[44] This would appear to be an underestimate, however, because Rubinshtein did not count transport workers (over 200,000) as well as wage labourers in small village industry. A more recent estimate by L.V. Milov puts the total number of hired labourers at about 0,5 million.[45] This estimate does not include hired labourers in services and trade, and therefore we revise Milov's estimate upward by 100.000, arriving at a total estimate of 600.000.

Bearing in mind that the absolute majority of wage workers were state serfs or nobles' serfs who paid the quitrent, we can incorporate Milov's estimate into the database. The combined number of 'state serfs' and 'Serfs (obrochnye)' is about 3 million, so for 1/6 of them

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[40] It should be noted, that the majority of servants also had their own households.

[41] Both those who paid the quitrent and those actually performing feudal labour obligations also carried out numerous other duties in providing the lord's estate with food, firewood, etc. The difference between quitrent and *barshchina* only relates to peasants' principal feudal obligations.

[42] Semevskii, V.I. *Krest'iane v tsarstvovanie imperatritsy Ekateriny II* (Moscow, 1903), p. 584.

[43] Kabuzan, *Narodonaselenie Rossii*, p. 122 and further.

[44] Rubinshtein, N.L., "Nekotorye voprosy formirovaniia rynka rabochei sily v Rossii v XVIII v.", *Voprosy istorii*, 1952, № 2, p. 95; See also: *Ocherki istorii SSSR. Period feodalizma. Rossiia vo vtoroi polovine XVIII v.* (Moscow, 1956), p. 112.

[45] Milov, L.V. ed., *Istoriia Rossii XVIII-XIX vekov* (Moscow, 2006), p. 245.

we add Free Wage Labour [Labour Relation 14018] as Lab_Rel 02.

7. Last but not least, the female population has to be added to the database. The overall balance between the sexes can be calculated on the basis of the figures for the female population as a whole, found by V.M. Kabuzan in the governors' reports[46], and it amounts to 1084 women per 1000 men[47]. Extrapolating this sex-ration to the database as a whole, we use this coefficient to calculate the number of women for each separate social group, with three important differences:

- there is no female group corresponding to the men serving in the army and navy
- women are always supposed to be kin producers within the household, and never leading household prodiucers (although in practice the opposite could sometimes have been the case)
- women are considered not to carry tributary labour obligations before the state, nor to be working as wage-earners, but the wives of '*barshinnye'* serfs in the nobles' estates are known to have been working at the lord's corvee fields and are therefore classified as Tributary Serfs alongside with their husbands.

1900

The principal source for the cross-section 1900 is the first-ever modern population census of the Russian Empire, held on 28 January 1897. The census turnouts have been published in 86 volumes for individual provinces (губернии) of the Empire, as well as in a two-volume national edition and a number of companion volumes with data on specific topics.[48] This database draws from (a) the data on occupations in table XX, published separately in four companion volumes, (b) a special two-part volume with the census data on workers, and (c) the data on the age-structure of the population in the national volumes.[49]

The database contains separate data for the sub-regions identified in the census as European Russia and Siberia. Together, this amounts to a territorial coverage consisting of the Russian Empire minus the Polish provinces, Finland (which did not participate in the census anyway), the Caucasus, and Central Asia.

Rather than relying on the data for European Russia and Siberia directly available from the census, we have chosen to aggregate these regional subtotals from province-level data available at the Electronic Repository of Russian Historical Statistics.[50] The main reason for this decision has to do with the comparability between data-sets, because there is a difference between the break-down of occupational data in the aggregate and the provincial census results. For reasons of economy, occupational data at the province level omit occupational groups which were present in insignificant numbers, and include the persons in these groups

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[46] Kabuzan, V.M. *Narody Rossii v XVIII v.: Chislennost' i etnicheskii sostav.* (Moscow, 1990), p. 183.
[47] The army was excluded from the calculations, as few soldiers had families.
[48] *Pervaia vseobschaia perepis' naseleniia Rossiiskoi imperii 28 ianvaria 1897 goda*, 86 vols. (St. Petersburg, 1905); *Obshchii svod po imperii rezul'tatov razrabotki dannykh pervoi vseobshchei perepisi naseleniia, proizvedennoi 28 ianvaria 1897 goda*, 2 vols. (St. Petersburg, 1905); Troinitskii, N.A. (ed), *Pervaia vseobshchaia perepis' naseleniia Rossiiskoi imperii 1897 g. Raspredelenie naselenie po vidam glavnykh zaniati i vozrastnym gruppam po otdel'nym territorial'nym raionam. Tablitsa XX* (St. Petersburg, 1905), 4 Vols.; Troinitskii, N.A. (ed), *Pervaia vseobshchaia perepis' naseleniia Rossiiskoi imperii 1897 g. Chislennost' i sostav rabochikh v Rossii na osnovanii dannykh Pervoi vseobshchei perepisi naseleniia Rossiiskoi imperii 1897 g.* (St. Petersburg, 1906), 2 Vols.
[49] *Obshchii svod [.] dannykh [.] perepisi 1897*, T. I, pp. 56-59; Troinitskii (ed), *Pervaia vseobshchaia perepis' naseleniia Rossiiskoi imperii 1897 g. Raspredelenie naselenie po vidam glavnykh zaniati i vozrastnym gruppam po otdel'nym territorial'nym raionam. Tablitsa XX*; Troinitskii (ed), *Pervaia vseobshchaia perepis' naseleniia Rossiiskoi imperii 1897 g. Chislennost' i sostav rabochikh v Rossii*
[50] Kessler, Gijs and Andrei Markevich (2019), Electronic Repository of Russian Historical Statistics, 18th - 21st centuries, https://ristat.org/

into an undifferentiated category 'other'. The data directly available from the census do not feature this category 'other', and are therefore essentially more precise, but because the numbers involved in this "other" category are quite small, we have chosen to give preference to considerations of comparability between the two available online-resources, and aggregate from the provincial data of the Electronic Repository of Russian Historical Statistics instead. Because the province-level data of the Electronic Repository of Russian Historical Statistics also rely on the Collaboratory taxonomy of labour relations, they complement our data-set, also because they cover all of the empire, including the territories not included here (Polish provinces, Caucasus, Central Asia).[51]

Separate break-downs by labour relation are provided for the male and the female population.

The first step in the creation of the database is to isolate the non-working part of the population, to be included in Labour Relation 1 - "Cannot work or cannot be expected to work". Using age-delimiters of ≤6 and ≥75 the population in these two age-categories are integrally allocated to Labour Relation 1.

Next, we turn to the census data on occupation from table XX. As said above, we use these data as available from the Electronic Repository of Russian Historical Statistics. Table XX in fact relates not so much to occupation, as to the source from which people derived their means of livelihood. At a first level it distinguishes between the people who had a source of income of their own on the one side, and the people who were supported by the former on the other side. For those who had a source of income of their own, this source of income is further specified, primarily by branch of industry or economic activity, but with some additional categories, like rentiers, prisoners, soldiers etc. (For a full list of the categories distinguished in table XX, cf. Appendix III) Some of these categories ought to be assigned to the category "non-working", like for example those who live on state support, those who are resident in shelters and institutions, the ill and the infirm, and others. Therefore, these are also classified as Labour Relation 1 in the database. This information relating to source of income is specified in the [Position] field.

Having thus isolated the non-working population, it is assumed next that all those listed in table XX as gainfully employed are above 6 and under 75 years of age. This is a simplification, because table XX itself reveals that relatively small numbers of the gainfully employed are outside of these age boundaries, but a necessary step to avoid double counting.

A further important implication of this assumption, is that the population between 6 and 75 years of age which is NOT listed as gainfully employed in table XX jointly makes up the Labour Relations 4b - "Household kin producers" [subordinate kin (men, women, and children) contributing to the maintenance of self-sufficient households], and 5 – "Household kin non-producers" [subordinate kin, including spouses (men and women) and children of heads of households, who can support the household (under either reciprocal or commodified labour relations)]. Ideally, one would of course be able to distinguish between Labour Relations 4b and 5, but unfortunately the data do not allow us to do so.

The next steps are to divide the people listed as gainfully employed in table XX over the Labour Relations 2-4 and 6-18. This cannot be done directly, due to the unfortunate

[51] As can be seen from Appendix IV, the occupational categories in Table XX have a two-tier structure, distinguishing between groups of occupational categories and a more refined second-tier level of precise occupational categories. The category "other" [Unspecified within group] emerges at the second level, and refers to the number of people who are known to have belonged to a particular group at the upper level of the hierarchy, but for whom no exact occupational titles are available at the second level. In some cases, when a group of occupational categories cannot be attributed to a particular labour relation in its entirety, but contains occupations which belong to different labour relations, this creates a problem, because it is not clear to which labour relations the "Unspecified within group" should be attributed. In such cases we have adhered to the principle that the "Unspecified" group follows the majority within the group.

circumstance that, for financial reasons, at the time the direct census data on labour relations [*положение в занятии*] were never worked out or aggregated. Only the data on blue-collar workers [*рабочие*] were published in a separate volume.[52] These are of limited value for our project, however, because blue-collar workers are only part of the larger categories 14 [market wage earners] and 18 [non-market wage-earners], next to white-collar workers. As a consequence, we have no choice but to use the data on occupation/source of income in table XX to infer the labour relations 2-18. This is feasible only to some extent. Separate data can be estimated/calculated for the Labour Relations 2, 4, 8, and 12-13-14-18:

Labour Relation 2 - "Affluent" [those who are so prosperous that they do not need to work for a living] is identified as such in table XX [sum of categories *14.01-Subsisting on income from capital and real estate* and *14.02-Subsisting on funds from parents or other relatives*].

Labour Relation 4a - "Leading household producers". This is an approximation. Russia was still an overwhelmingly peasant society in 1897, with some inroads of more market-oriented agricultural production in certain regions. Logically, we can therefore expect the population gainfully employed in agriculture to have consisted of four categories, in terms of labour relations: heads of peasant households [mainly 4a – Leading producers, but probably also some 12a – Self-employed leading producers], non-heads of household participating in agricultural work [4b – Kin producers], agricultural labourers [14 – Wage earners], and their employers [either 13-Employers, or 4a – Leading producers and 12a – Self-employed leading producers, in as far as agricultural labourers were hired for work by peasant households]. How to separate these from the available data?

Table XX differentiates between *17-Agriculture, 18-Apiculture and sericulture,* and *19-Livestock farming*, with a further break-down within these larger groups, among which *17.01-Crop farming overall*, *19.01-Livestock farming overall* and several other specialised branches of agriculture and animal husbandry. The number of agricultural hired labourers working in agriculture is available from the separate set of volumes on blue-collar workers.[53] Unfortunately, it does not distinguish any further between the occupational groups 17, 18 and 19. Therefore we have no choice but to amalgamate these three categories into a combined *17_18_19-Agriculture; Apiculture & sericulture; Livestock farming*. Subtracting the number of agricultural labourers from the total number of people working in agriculture we are left, according to our scheme as set out above, with a combined category 4a-4b-12a-12b-13.

However, it has been noticed in the literature, that the 1897 census under-enumerates employment in agriculture, because it only appears to have counted heads of households, and not kin producers.[54] This is of course a defect of the census, but it can be turned to our advantage here. Because, if true, it means that kin producers 4b and 12b are not included in the totals on agricultural employment, which leaves us with a combined category of 4a-12a-13. There is no real possibility to separate these three categories in a meaningful way, but because the agricultural sector was only just embarking on a more market-oriented course, we assume that the overwhelming majority of employers in the agricultural sector, must have been peasant households who resorted to the use of additional hired labour at some point in the agricultural year-cycle.[55]

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[52] Troinitskii (ed), *Pervaia vseobshchaia perepis' naseleniia Rossiiskoi imperii 1897 g. Chislennost' i sostav rabochikh v Rossii,* foreword.

[53] Troinitskii (ed), *Pervaia vseobshchaia perepis' naseleniia Rossiiskoi imperii 1897 g. Chislennost' i sostav rabochikh v Rossii*

[54] Gozulov A.I., *Perepisi naseleniya SSSR i kapitalisticheskikh stran (opit istoriko-metodologicheskoj kharakteristiki proizvodsvta perepisej naseleniya)*. (Moscow, 1936), pp. 205-206;

[55] The idea that peasants do not hire labour from outside, is tenacious, but has long since been disproven. In fact, peasants often resort to the use of hired labour on an habitual basis, without this necessarily meaning that they have departed from an economic logic of subistence-oriented family smallholder farming. Cf. Kessler, Gijs, "Wage Labor and the Household Economy: a Russian Perspective, 1600-2000", in Marcel Van der Linden and

This assumption brings us to the point where we can simply split the total employment in agriculture into two parts: (a) the number of agricultural labourers as specified in the census [set aside as Labour Relation 14 – Wage workers] and heads of peasant households [Labour Relation 4a].[56]

Labour Relation 6 - "Household servants" and Labour Relation 7 - "Community-based redistribution agents" are non-existant for the benchmark 1900.

Labour Relation 8 - "Forced Labourers" can be calculated quite precisely from table XX as the sum of (a) men between 6 and 75 years of age serving in the armed forces (army and navy) as non-commanding officers and (b) convict labourers.

Labour Relations 9, 10 and 11 are non-existent for the benchmark 1900.

The biggest single shortcoming of the data in table XX is that they do not allow us to separate the Labour Relations 12a [Self-employed leading producers], 12b [Self-employed kin producers], 13 [Employers], 14 [Market Wage-earners] and 18 [Wage-earners employed by non-market institutions]. Table XX merely gives us the number of people who derived their income from a particular branch of production, trade or services, whether in employment, self-employment, or indeed as employer. Also, it is problematic to separate between wage-earners working for the market [Labour Relation 14] and wage-earners employed by non-market institutions [Labour Relation 18]. This is because the data are ordered by occupation or type of production, rather than by the ownership of the branch concerned. Particularly in manufacturing the distinction between those employed by private companies and those employed by state enterprises or belonging to a co-operative is impossible to draw on the basis of table XX. Therefore, although for some occupations and/or trades it might be feasible to build valid assumptions as to the ownership structure of the establishment they were plying their trade at, the aggregate picture would be distorted and for this reason we have chosen not to separate the Labour Relations 14 and 18 in the database.

Together, these two limitations yield an amalgamate category 12a-12b-13-14-18, comprising the upper-level category "commodified labour" in its entirety. These data are broken down by branch and occupation, using the categories from the census, translated into English for the purpose (for the original Russian terms, please refer to the lookup-table in Appendix IV). Note that for each group this contains a category "Other" ["Unspecified within group"], the presence of which is due to the circumstance that these data were aggregated from the province level data, where a full breakdown is not always available (see above).

In addition to the main data-set for 1900 [Russia_1900_(OH&GK-April2019).xlsx], we have undertaken an attempt to split up this lump category "commodified labour" into its constituent parts. This procedure relies on a number of assumptions and additional data, and in order to keep the main data-set as clean as possible we have chosen to provide the estimates thus produced as an alternative version of the data-set [Russia_1900_(OH&GK-Sept2019).xlsx]. The estimation procedures and the estimates thus produced are treated in detail in Appendix III to this methodological paper.

Leo Lucassen (eds), *Working on Labor. Essays in Honor of Jan Lucassen*, Studies in Global Social History, 9 (Leiden [etc.], 2012), pp. 355-357.

[56] Note, that some agricultural hired labourers will also have been heads of household, so there is some overlap between the categories, but we will ignore this.

2000

The data-set for the benchmark year 2000 is based on the first population census of the post-soviet Russian Federation, held on 9 October 2002. Data for other Soviet successor states are not part of the data-set, but are available separately. Like the Russian data, they are drawn from the population censuses held in these countries as part of the so-called 2000-round, and relate to the following dates: Belarus[57] [16 February 1999], Ukraine [5 December 2001, forthcoming], Estonia [31 March 2001, forthcoming], Latvia [31 March 2001, forthcoming], Lithuania [6 April 2001, forthcoming], Moldova [5-12 October 2004, forthcoming]. The data for the individual countries can be used in their own right or aggregated in order to obtain a territorial coverage comparable to that of the 1800 and 1900 data-sets.

Because of the different dates on which the population censuses in the respective countries have been conducted, migration between the countries obviously distorts the picture to some extent, in particular labour migration from East Ukraine, Moldova, and Belarus to the Russian Federation, for work in construction and other sectors of the economy. As most of these temporary migrants would, however, have avoided participation in the 2002 census in the Russian Federation, due to their uncertain or illegal employment, fiscal and immigration status, double-counting might have been much less significant than migration figures would suggest. In any case, there is no conceivable way of adequately correcting the figures for double-counting and therefore the distortion can only be accepted as given.

The paragraphs below specify the procedures which have been followed in compiling the data-set for the Russian Federation from the tables of the 2002 census turnouts:[58]

- volume 2, table 1 [Все население, городское, сельское по полу и возрасту = Rural and urban population by age and sex]
- volume 5, table 3 [Население, указавшее один источник средств к существованию, по полу и возрастным группам = Population with one source of income by age and sex]
- volume 6, table 12 [Население коллективных домохозяйств по полу и возрастным группам = Members of institutional households by age and sex]
- volume 7, table 1 [Население частных домохозяйств в возрасте 15-64 лет по возрастным группам и экономической активности = Non-institutional household population between 15 and 64 by age and economic activity]
- volume 7, table 5 [Занятое в экономике население в возрасте 15 лет и более по возрастным группам и положению в занятии = Employed population over 15 years of age by age and labour relation]
- volume 8, table 8 [Занятое население в возрасте 15-64 лет по видам экономической деятельности и положению в занятии = Employed population between 15 and 64 years of age by type of economic activity and labour relation]

As can be seen from this list, some of the tables relate to the population as a whole, whereas others relate only to the members of multi-person households or the population of working age. This requires a number of assumptions in our calculations, which will be specified below. The census data allow us to identify the Labour Relations 1, 2, 3, 4a, 4b/5, 12a/b, 13, 14/18. All data on labour relations are broken down by sex, and, where possible, by age.

Starting with Labour Relation 1 - “Cannot work or cannot be expected to work” the age-

[57] Hofland, Olav, 2018, "Belarus 2000 [Global Collaboratory on the History of Labour Relations 1500-2000 Dataset]", http://hdl.handle.net/10622/UZ5ULR, IISH Dataverse, V1

[58] *Itogi vserossiiskoi perepisi naseleniia 2002 g.*, 14 vols. (Moskva, 2004-2005)

delimiters applied for separating those too young or too old to work or to be expected to work from the rest of the population are 15 and 74 years of age, i.e. all men and women under 15 and over 74 [Vol. 2, Table 1] are integrally assigned to Labour Relation 1. The lower age-delimiter reflects the existence of obligatory school-education and is generally adhered to in current labour and employment statistics. The upper age-delimiter is more controversial - the Russian Federation inherited the Soviet pension system introduced in the late 1950s, with ages of retirement set at 60 for men and 55 for women. However, whereas in Soviet times this meant actual retirement at these ages, the decline of the value of retirement pensions in real terms during the post-Soviet period has effectively transformed the pension into a monetary allowance which is received irrespective of employment status. In addition to the pension people would have to provide for the means of existence, or be supported by relatives. Considering the definitions of work adhered to in our taxonomy, which includes household work, only when no contribution whatsoever would be expected anymore from them, should they be classified as belonging to Labour Relation 1. The default value for this age in the taxonomy is 75 and there is no reason not to consider this an appropriate value for Russian society at the start of the twenty-first century as well.

Further categories to be included in Labour Relation 1 are identified as such in the census:

(a) all people over 15 years of age living in institutional households [Vol. 6, table 12]. Apart from mental homes, hospitals, orphanages etc. this also includes the prison population, the armed forces living in barracks, as well as monasteries and convents. The latter three subcategories are included in the undifferentiated category "other institutions/другие учреждения".

(b) people listed in Vol. 7, table 1 as “people receiving social welfare (except unemployment benefits) and people on state support” [лица, получающие пособие (кроме пособия по безработице) и лица, находящиеся на обеспечении государства], and “grantees” [стипендиаты]. These two categories would include most students, as well as the disabled. Unfortunately, though, these data only relate to those of 15-64 years of age, whereas in our taxonomy we would also want to include those of 65-74 years of age. As far as students are concerned, this is a minor omission, but relative to the disabled and other people on state support it has to be acknowledged that our data contain a certain downward bias.

Labour Relation 2 - “Affluent” [those who are so prosperous that they do not need to work for a living] has been calculated as the number of people over 15 years of age reporting savings [сбережения] or the letting of property [доход от сдачи внаем или в аренду имущества] as their single source of income [Vol. 5, table 3]. This contains a slight upward bias in that it includes all people over 65 years of age, i.e. inclusive of those over 74, whom we have already integrally allocated to Labour Relation 1 and therefore introduces a certain double-counting. To compensate this to some extent the number of people in the table who had indicated no age have been excluded from the calculations.

Labour Relation 3 - “Unemployed”. The census only specifies the number of unemployed for the population between 15 and 64 living in non-institutional households [Vol. 7, table 1]; it is assumed that among the remaining categories of the population (people in institutions, the homeless and the non-institutional household population between 64 and 75) the number of unemployed will have been negligible.

Labour Relation 4a - “Leading household producers”. Calculated as the number of rural residents working on their own account in agriculture, hunting, fishing, forestry and aquaculture [*Сельское население, работающие не по найму без привлечения наемных*

работников в с-х, охоте, лесном хозяйстве, рыболовстве и рыбоводстве]. This figure only concerns the non-institutional household population between 15 and 64 [Vol. 8, table 8]. Of course outside non-institutional households few self-sufficient household producers are to be found, with the exception of heads of monasteries, but the omission of the age-category 65-74 is more problematic - it is exactly among the elderly that a lot of the self-sufficient rural producers can be found. On the other hand, the data overestimate the number of self-sufficient households, as some of the rural producers not working with hired labour should in fact be classified as self-employed, because they might be primarily working for the market. Therefore, on average, this is probably as good an estimate of the number of leading household producers in the countryside as we can get on the basis of available data.

Labour Relation 4b5 - "Household kin producers; Household kin non-producers". Calculated as a residual category, obtained by subtracting Labour Relations 1, 2, 3, 4a, 12a/b, 13, 14/18 from the total population. This figure is inclusive of women between 55 and 75 and men between 60 and 75 who are formally retired and in receipt of retirement pensions, but are not specifically gainfully employed. As we have argued above, pensions are not adequate for covering living expenses and people of this age would therefore either supplement them with income from gainful employment, or would be supported by relatives, taking care of grandchildren or performing other household duties in return.

People with an unspecified age or with unspecified employment were also included in the amalgamate, residual category 4b/5.

Unfortunately, the census data do not allow us to identify military conscripts and therefore, Labour Relation 8 - "Forced Labourers". They cannot be separated from professional army personnel and have therefore been included with Labour Relation 18.

Labour Relations 12ab, 13, and 1418 are calculated on the basis of the full data on employment in Vol. 7, table 5. This causes a certain double-counting and overlap with Labour Relation 1: Volume 7, table 5 relates to the population over 15 years of age, i.e. inclusive of the population above 74, which we have integrally allotted to Labour Relation 1. Although a breakdown of the data according to age-cohorts is provided in the table, it does not provide separate figures for those over 74 years of age, but lumps them together with the age-cohort 64-74. Therefore we have no choice but to accept the possible distortion, also because the number of people over 74 years of age who would have indicated in the census to have been gainfully employed, must have been quite small.

Labour Relations 12a - "Self-employed leading producers" and 12b – "Self-employed kin producers". Because of the often informal character of self-employment we have opted to calculate this category as a residual category, including everyone who indicated to be gainfully employed, but NOT to be working in employment. This involves adding up the Vol. 7, table 5 categories "working in self-employment without hired personnel" [*работающие не по найму без привлечения наемных работников*], "working in self-employment, without indication of the presence of hired labour" [*работающие не по найму, не указавшие привлечение наемных работников*] and "working in self-employment - other" [*работающие не по найму - иное*], as well as the top-level category of those indicating gainful employment in general, without further specifying their labour relation. To avoid double-counting we subsequently subtract the rural self-employed already classified as Labour Relation 4a from this total. The data do not allow us to distinguish between self-employed leading producers and self-employed kin producers, therefore we include them as the amalgamate category 12a/b.

Labour Relation 13 - "Employers" are identified as such in Vol. 7, table 5 [*работающие не по найму с привлечением наемных работников*].

Labour Relation 14 [Market Wage-earners] and 18 [Wage-earners employed by non-market institutions] cannot be distinguished from each other on the basis of the census-data and are therefore calculated as an aggregate category, comprising all those listed in Vol. 7,

table 5 as working in employment [*работающие по найму*].

The database contains only the upper-level data, providing the breakdown of the population into the labour relations enumerated above for men and women separately. Unfortunately, they cannot be differentiated any further by branch, trade and occupation because such a breakdown is available only for the population of 15-64 years of age as listed in Vol. 8, table 8. For reasons unspecified in the census volumes these data are not compatible with the data on labour relations from Vol. 7, table 5, which include all of the population over 15 years of age and, for that reason, have been used for the compilation of our database. The incompatibility lies in the fact that a simple subtraction of the number of people in the age-category ≥ 65 from the total in Vol. 7, table 5, which in theory should yield the same total as that listed in Vol. 8, table 8 for the population of 15-64, nonetheless returns a figure of over one million persons more, a difference in the order of magnitude which cannot be just ignored. For that reason we have decided to abstain from a classification by trade or occupation, also because the breakdown provided in the census is only a rudimentary one by sector of the economy.

All labels and categories from the census used in compiling this data-set are included in English translation in the Position field of the database. For the original Russian terms the user is referred to the look-up table in Appendix IV below.

Overview of the Data

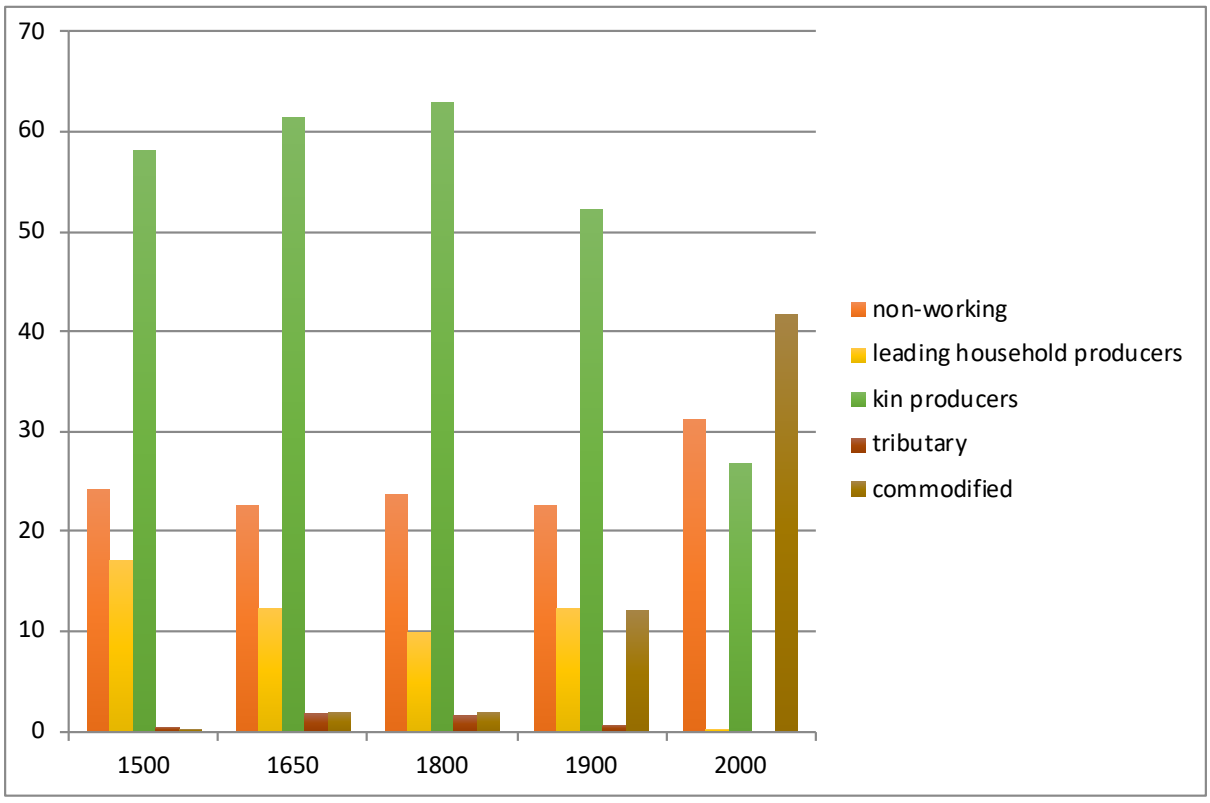
a) Labour relations and combinations of labour relations identified in the databases in percentages of the population:

LabRelation old	LabRelation new	1500	1650	1800	1900	2000
1	1	109,290	2,232,090	9,142,534	21,997,887	37,794,506
2	2		40,950	11,182	437,749	178,187
3	3		230			7,411,186
4	4a	71,806	297,654	713,659	12,162,729	130,344
4+ smth else	4a+ smth else	5,750	932,807	3,122,274		
5	4b5	255,806	2,356,596	14,219,953	51,861,799	38,998,103
5+ smth else	4b5+ smth else	7,881	3,826,839	10,152,201		
6	6		666			
6+ smth else	6+ smth else		2			
8	8	1,848	40,950	596,306	694,286	
10	10	533	134,178			
10+smth else	10+smth else		53			
12	12a/b	209	954			1,425,966
12+smth else	12a/b+smth else		194,329			
13	13		87			924,185
13+smth else	13+smth else					
12 AND/OR 13 AND/OR smth else	12a/b AND/OR 13 + smth else			160,423		
12 AND/OR 13 AND/OR 14	12a/b AND/OR 13 AND/OR 14/18				10,111,777	
14	14		10		1,887,241	
14 AND/OR 18	14 AND/OR 18				48,219	58,304,254
18	18		1,799	5,000		
18+smth else	18+smth else			10,651		
12 AND/OR 13 AND/OR 14 AND/OR 18+ smth else	12a/b AND/OR 13 AND/OR 14 AND/OR 18+ smth else			602,473		
TOTAL POPULATION	TOTAL POPULATION	453,123	10,060,194	38,736,656	99,201,687	145,166,731

b) Labour relations and combinations of labour relations according to main categories of taxonomy in percentages of the population:

LabRelation	Category	1500	1650	1800	1900	2000
1 OR 2 OR 3	Non-working	24.1	22.6	23.6	22.6	31.3
4 OR 4+ smth else	Household producers	17.1	12.2	9.9	12.3	0.1
5 OR 6 AND 5 OR 6 + smth else	Kin producers	58.2	61.5	62.9	52.3	26.9
8 OR 10 AND 8 OR 10 + smth else	Tributary labour	0.5	1.7	1.5	0.7	0.0
12 OR 13 OR 14 OR 18 AND 12/13/14/18 + smth else	Commodified labour	0.0	2.0	2.0	12.1	41.8
Total population		100.0	100.0	100.0	100.0	100.0

c) Percentages of total population by main categories taxonomy:



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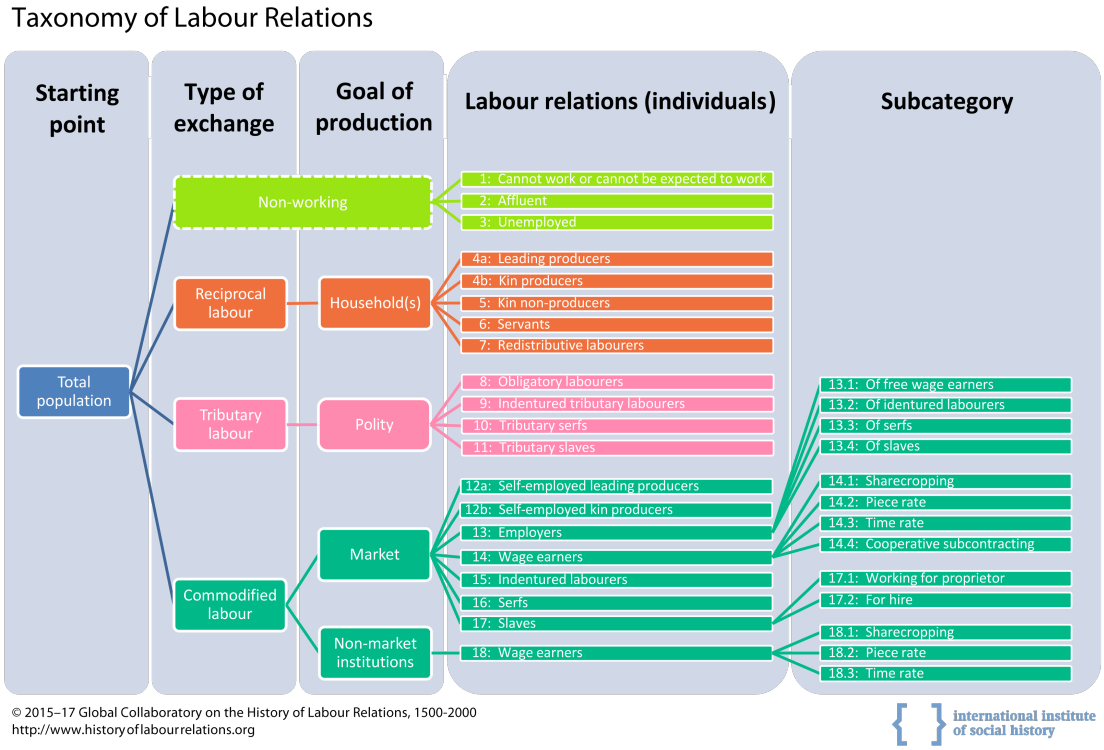
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Appendix I – Taxonomy and definitions of Labour Relations

[NEW: databases 1900 & 2000]



Definitions of Labour Relations

Non-working:
As a starting point for each geographical unit and cross section, we take the entire population and subsequently determine what part is not, as a rule, working, and, consequently, what part is working (these "calculations" will often be based on estimates rather than precise data). The non-working population is divided into the following three categories:

1. *Cannot work or cannot be expected to work:* those who cannot work, because they are too young (≤6 years), too old (≥75 years),[59] disabled, or are studying.
2. *Affluent*: those who are so prosperous that they do not need to work for a living (rentiers, etc.), and consequently actually do not work. This also goes for their spouses if all their productive and reproductive tasks are taken over by servants, nannies, etc. There are, of course, affluent people, owners of big companies, who are wealthy enough to stop working but nevertheless choose to continue to work. If they are employers, these people should be assigned to labour relation 13 instead of 2.
3. *Unemployed*: although unemployment is very much a nineteenth- and, especially, twentieth-century concept, we do distinguish between those in employment and those wanting to work but who cannot find employment.

Working:
Reciprocal labour:
Persons who provide labour for other members of the same household and/or community are

[59] These minimum and maximum ages are very much culturally determined. The age brackets chosen will always be indicated in the database and explained in the methodological paper.

subsumed within the category **Reciprocal labour**.

Within the household:

4 a *Leading household producers*: heads of self-sufficient households (these include family-based and non-kin-based forms). Self-subsistence can include small market transactions, but only if most (at least eighty per cent) of total household income is earned through self-subsistence labour. Heads of households have labour relation 4a.

4 b *Household kin producers:* subordinate kin, including spouses (men and women) and children of the above heads of households, who are mainly self-subsistent and who contribute to the maintenance of the household by performing productive work for that household.

5. *Household kin non-producers:* subordinate kin, including spouses (men and women) and children of heads of households, who can support the household (under either reciprocal or commodified labour relations). These spouse and kin dependants are free from productive work, but they contribute to the maintenance of the household by performing reproductive work for the household, i.e. especially child rearing, cooking, cleaning, and other household chores.
   In all other cases spouses and kin producers in the categories named have income-generating activities essential for the survival of the household, i.e. labour relations 12a, 12b, 13, 14, or 18, and will have one of these labour relations themselves.
6. *Reciprocal household servants and slaves*: subordinate non-kin (men, women, and children) contributing to the maintenance of self-sufficient households. This category does not include household servants who earn a salary and are free to leave their employer of their own volition (i.e. labour relation 14), but it does include servants in autarchic households, monasteries, and palaces. They may work under all shades of conditions, from enforcement (including pawnship) to a desire to receive patronage. These conditions may change from one generation to another.[60]

Within the community:

7. *Community-based redistributive labourers*: persons who perform tasks for the local community in exchange for communally provided remuneration in kind, such as food, accommodation, and services, or a plot of land and seed to grow food on their own. Examples of this type of labour include working under the Indian *jajmani* system, hunting and defence by Taiwanese aborigines, or communal work among nomadic and sedentary tribes in the Middle East and Africa. In the case of the *jajmani* workers in South Asia, hereditary structures form the basis of the engagement, while in parts of Africa or Taiwan the criteria for fulfilling community-based labour are gender and age (in Taiwan, for example, males between six and forty).

Tributary labour:

Persons who are obliged to work for the polity (often the state, though it could also be a feudal or religious authority). Their labour is not commodified but belongs to the polity. Those workers are included in the category **Tributary labour**.

8. *Obligatory labourers*: those who have to work for the polity, and are remunerated mainly in kind. This category includes those subject to civil obligations (corvée labourers, conscripted soldiers and sailors), and work as punishment, i.e. convicts. Yet the obligatory work can also be an entitlement that enjoys middle or high social standing, such as the European or Indian nobility, the samurai in Japan, or banner people in Qing China.
9. *Indentured tributary labourers*: those contracted to work as unfree labourers for the polity for a specific period of time to pay off a debt or fine to that same polity.
10. *Tributary serfs*: those working for the polity because they are bound to its soil and bound to provide specified tasks for a specified maximum number of days, for example state serfs in Russia.
11. *Tributary slaves*: those who are owned by and work for the polity indefinitely (deprived of the right to leave, to refuse to work, or to receive compensation for their labour). One example is forced labourers in concentration camps.

[60] First-generation slaves might be commodified, whereas their children might no longer be considered slaves and might be working for the household on conditions that were more free, as was the case with children of slave women and free Ashanti men. See Gareth Austin, *Labour, Land and Capital in Ghana: From Slavery to Free Labour in Asante, 1807-1956* (Rochester, NY, BOYE6, 2005), pp. 106-134, 174-180, 481-490, 498-500.

Commodified labour:
Work done on the basis of market exchange in which labour is "commodified", i.e. where the worker or the products of his work are sold. The category **Commodified labour** is subdivided into those working for the market and those working for public institutions *that may nevertheless produce for the market* (though not for the gain of private individuals).

For the market, private employment:

12 a *Self-employed leading producers*: those who produce goods or services for the market (for example, peasants, craftsmen, petty traders, transporters, as well as those in a profession) with fewer than three employees, possibly in cooperation with

12 b *Self-employed kin producers:* household members including spouses and children who work together with self-employed leading producers who produce for the market. All members of a family working under a putting-out system should be counted as self-employed producers.[61]

13. *Employers*: those who produce goods or services for market institutions by employing more than three labourers. The number after the dot is an attribute that says something about the freedom or unfreedom of the employees.
    - 13.1 Employers who employ free wage earners.
    - 13.2 Employers who employ indentured labourers.
    - 13.3 Employers who employ serfs.
    - 13.4 Employers who employ slaves.
14. *Market wage earners*: wage earners (including the temporarily unemployed) who produce commodities or services for the market in exchange mainly for monetary remuneration. A subdivision is made by type of remuneration.
    - 14.1 Sharecropping wage earners: remuneration is a fixed share of total output.
    - 14.2 Piece-rate wage earners: remuneration at piece rates.
    - 14.3 Time-rate wage earners: remuneration at time rates
    - 14.4 Cooperative subcontracting workers at piece rates.
15. *Indentured labourers for the market*: those contracted to work as unfree labourers for an employer for a specific period of time to pay off a private debt. They include indentured European labourers in the Caribbean in the seventeenth and eighteenth centuries, and indentured Indian, Chinese, and Japanese workers after the abolition of slavery.
16. *Serfs working for the market*: those bound to the soil and bound to provide specified tasks for a specified maximum number of days for private landowners, for example serfs working on the estates of the nobility.
17. *Slaves who produce for the market*: those owned by their employers (masters). They are deprived of the right to leave, to refuse to work, or to receive compensation for their labour. Here we do not distinguish between the different ways individuals may become enslaved (sale, pawning, etc.). We do, however, differentiate between:
    - 17.1 Slaves working directly for their proprietor, for example productive work by plantation slaves, and domestic slavery in households producing for the market.
    - 17.2 Slaves for hire, for example for agricultural or domestic labour (as a rule they may keep a small part of their earnings, while the largest part goes to the owner).

For non-market institutions:

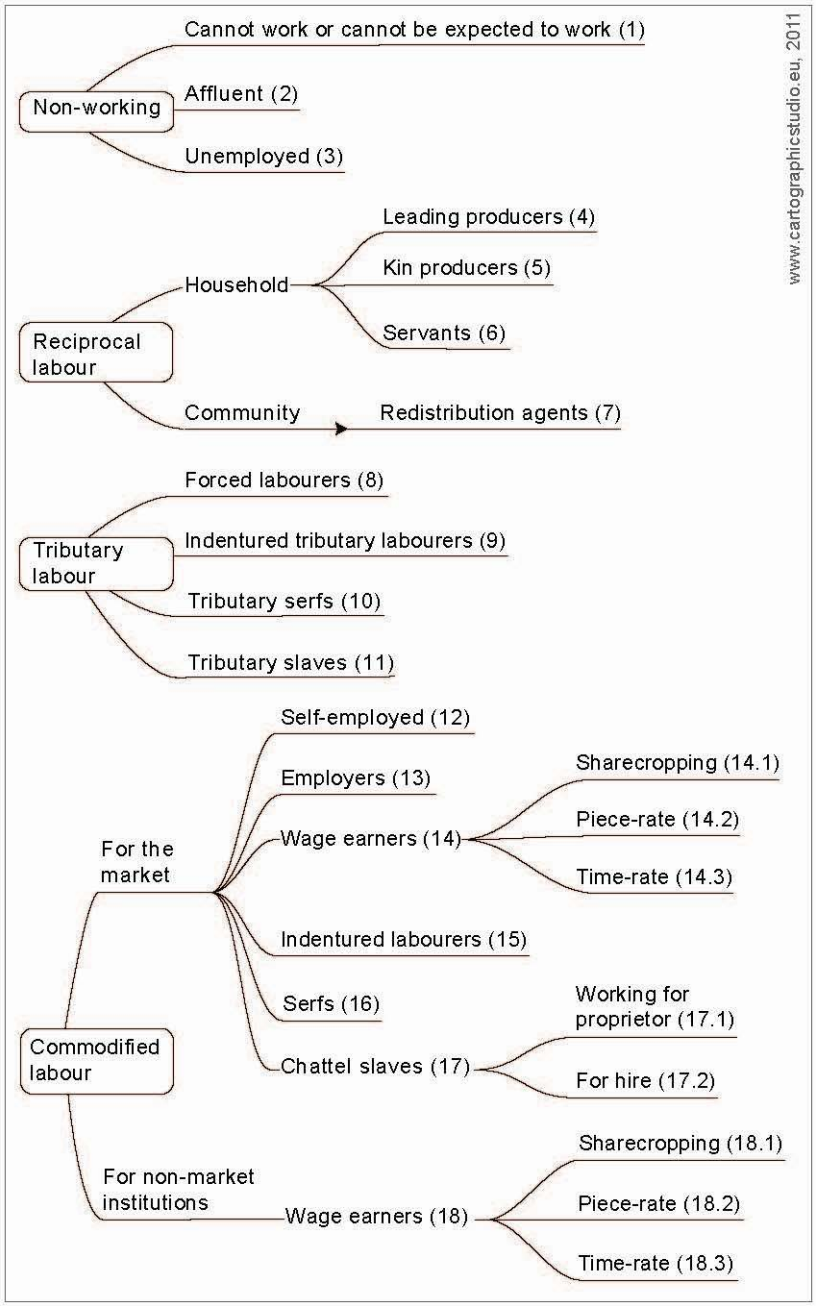
18. *Wage earners employed by non-market institutions* (that may or may not produce for the market), such as the state, state-owned companies, the Church, or production cooperatives, who produce or render services for a free or a regulated market. A subdivision is made by type of remuneration:
    - 18.1 Sharecropping wage earners: remuneration is a fixed share of total output.
    - 18.2 Piece-rate wage earners: remuneration at piece rates.
    - 18.3 Time-rate wage earners: remuneration at time rates.

[61] As long as they are ≥ 6 and ≤ 75 (or other age indications for too young or too old to work as documented for the cross section that is specifically analysed).

Appendix II – Taxonomy of Labour Relations

[OLD: databases 1500-1650-1800]

The taxonomy of labour relations, showing types of labour relations (labrels) 1 to 18 in brackets



Definitions of Labour Relations
As a starting point for each geographical unit and cross-section, we take the whole population, and determine which part is as a rule not working, and, consequently, what part is working (these calculations are often based on estimates, rather than on direct data). The non-working population is divided into three categories:

- Those **who cannot work or cannot be expected to work** (because they are either too young, too old, disabled or studying).

- Those who do not have to work (because they are **affluent**).

- Those who wish to work but have no job (**unemployed**).

A central focus of the Collaboratory is the possible impact of the market economy. To help analyze that influence, the working population is also divided into three categories:

- Workers who provide labour within the household and the community are subsumed within the category **reciprocal labour**.
- Others are obliged to work for the polity (often the state, though it could also be a feudal or religious authority). Their labour is not commodified, and is owned by the polity. Those workers are included in the category **tributary labour**.
- One of the main trends we examine is the commodification of labour. The employer who 'buys' labour power may produce for the market, or for non-market institutions. Therefore, the category **commodified labour** is subdivided into those working for the market, and those working for public institutions *which may nevertheless produce for the market* (though not for the gain of private individuals). In some statistical sources, those comprising the category of commodified labour are referred to as 'gainful workers'.

Combining all these categories, we arrive at the following taxonomy of labour relations (labrels):

Non-working:

4. *Cannot work or cannot be expected to work:* those who cannot work, because they are too young (≤6 years), too old (≥75 years),[62] disabled, or are studying.
5. *Affluent*: those who are sufficiently prosperous, so that they do not need to work for a living (rentiers, etc.).
6. *Unemployed*: although 'unemployment' is very much a nineteenth-century (and, especially, twentieth-century) concept, we do distinguish between those in employment, and those wishing to work but who cannot find employment.

Working:

Reciprocal labour:

Within the household:

7. *Leading household producers*: heads of self-sufficient households (these include family-based and non-kin-based forms, such as monasteries and palaces). In many households after 1500, 'self-sufficiency' can no longer have been complete. Basic foodstuffs (e.g. salt), and materials for tools and weapons (e.g. iron), were acquired through barter or monetary transactions – even in tribal societies which were, by 1500, only marginally exposed to market production.[63] 'Self-sufficiency' in our sense, which occurs in labour relations 4, 5, and 6, can include small-scale market

[62] These minimum and maximum ages are very much culturally determined. If these differ greatly for a certain region or a certain cross-section, this difference is indicated in the database.
[63] According to Amalendu Guha, "The Medieval Economy of Assam", in Tapan Raychaudhuri and Irfan Habib (eds), *The Cambridge Economic History of India* (Cambridge, 1982), vol. 1, p. 487, "village self-sufficiency in a total sense was a myth", even for the relatively remote sixteenth- and seventeenth-century Assam.

transactions that aim at sustaining households, rather than accumulating capital by way of profiting from exchange-value.[64]

8. *Household kin producers*: subordinate kin (men, women, and children) contributing to the maintenance of self-sufficient households.
9. *Household servants*: subordinate kin (men, women, and children) contributing to the maintenance of self-sufficient households. This category does not refer to household servants who earn a salary, and who are free to leave their employer of their own volition, but instead to servants in feudal autarchic households.

Within the community:

10. *Community-based redistribution agents*: persons who perform tasks for the local community, in exchange for communally provided remuneration in kind – such as food, accommodation, and services, or a plot of land and seed to grow food on their own. Examples of this type of labour include work under the Indian *jajmani* system, hunting and defence by Taiwanese aborigines, or communal work in nomadic and sedentary tribes in the Middle East and North Africa. In the case of the *jajmani* workers, hereditary structures form the basis of the engagement, while in Taiwan the criteria for fulfilling community-based labour are gender (male) and age (between the ages of six and forty).

Tributary labour:

11. *Forced labourers*: those who have to work for the polity, and are remunerated mainly in kind. They include corvée labourers, conscripted soldiers and sailors, and convicts.
12. *Indentured tributary labourers*: those contracted to work as unfree labourers for the polity, for a specific period of time, to pay off a debt. For example, German regiments (the 'Hessians') in the service of the British Empire, who fought against the American colonists during the American Revolutionary War.
13. *Tributary serfs*: those working for the polity, because they are bound to the land owned by the polity, and compelled to perform specified tasks for a specified maximum number of days.
14. *Tributary slaves*: those who are owned by, and work for the polity indefinitely (deprived of the right to leave, to refuse to work, or to receive compensation for their labour). Forced labourers in concentration camps are an example.

Commodified labour:

For the market, private employment:

15. *Self-employed*: those who produce goods or services for market institutions, possibly in cooperation with other household members or no more than three wage labourers, apprentices, serfs, or slaves (for example, peasants, craftsmen, petty traders, transporters, as well as those in a profession). *Nota bene*: All members of a family working under a putting-out system are counted as self-employed producers.[65]
16. *Employers*: those who produce goods or services for market institutions by employing more than three wage labourers, indentured labourers, serfs, or slaves.
17. *Market wage earners*: wage earners who produce commodities or services for the market, in exchange mainly for monetary remuneration.

    14.1. Sharecropping wage earners: remuneration is a fixed share of total output (including temporarily unemployed persons).

    14.2. Piece-rate wage earners: remuneration at piece rates (including temporarily unemployed persons).

    14.3. Time-rate wage earners: remuneration at time rates (including temporarily unemployed persons).

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[64] Marcel van der Linden, "Global Labour History and 'the Modern World-System'", *International Review of Social History*, 46 (2001), pp. 423-459, p. 452, referring to G.A. Cohen, *Karl Marx's Theory of History: A Defence* (Oxford, 1978). See too Akira Hayami, Osamu Saitô, and Ronald P. Toby (eds.), *The Economic History of Japan, 1600-1990,* Vol. 1: *Emergence of Economic Society in Japan, 1600-1859* (Oxford, 2004), who argue in a similar sense for a broader concept of "self-sufficiency", which allows for market production as long as accumulation is not the basic goal of such transactions.

[65] As long as they are ≥ 6 and ≤ 75.

18. *Indentured labourers for the market*: those contracted to work as unfree labourers for an employer for a specific period of time, to pay off a debt. They include indentured labourers in the British Empire after the abolition of slavery.
19. *Serfs working for the market*: those bound to the land and compelled to perform specified tasks for a specified maximum number of days (for example, the serfs working on large grain-growing estates of the nobility east of the Elbe).
20. *Chattel slaves who produce for the market*: those owned by their employers (masters). They are deprived of the right to leave, the right to refuse to work, and/or the right to receive compensation for their labour.

    17.1 Sharecropping chattel slaves working for their proprietor (for example, plantation slaves working in the Caribbean).

    17.2 Slaves for hire (for example, agricultural or domestic labour in eighteenth-century Virginia).

For non-market institutions that may produce for the market:

21. *Wage earners employed by non-market institutions,* such as the state, state-owned companies, the Church, or production cooperatives, who produce or render services for a free or a regulated market.

    18.1 *Sharecropping wage earners*: remuneration is a fixed share of total output (including the temporarily unemployed).

    18.2 *Piece-rate wage earners*: receiving remuneration at piece rates, including temporarily unemployed (for example, hired artisans in Chinese imperial silk weaving enterprises during the Ming and Qing dynasties).

    18.3 *Time-rate wage earners*: remuneration at time rates, (for example, hired artisans on Chinese imperial construction projects during the Ming and Qing dynasties, and workers and employees in twentieth-century state enterprises).

Appendix III – Estimating Commodified Labour Relations, 1900

Unfortunately, the 1897 census only provides the total number of people working in the different branches of production, without distinguishing between employers, the employed, and the self-employed. This is a great pity, particularly because this extra information was in fact gathered by the census-takers. But it was excluded from the subsequent aggregations for budgetary reasons.[66] In the main data-set we have therefore resorted to the use of an 'amalgamate' category 12a-13-14-18, or "Commodified Labour" according to the collab taxonomy.

However, making a number of assumptions, and drawing in some extra data, informed estimates can be made for the relative shares of the constituent labour relations within the larger category "Commodified labour". This appendix sets out the procedure followed to arrive at these estimates. Because of the number of assumptions involved we have deliberately chosen not to make the estimates a fully integrated part of the data-set, but to provide them as an alternative data-set.

Crucially, one year after the main body of census results, two additional volumes appeared with direct data on the number of manual workers (*rabochie*), published in order to salvage at least part of the data on labour relations.[67] Subtracting manual workers from the total number of people in "commodified labour" we are left with a residual category, consisting of employers, the self-employed and white-collar workers. At the level of the total workforce, this residual category cannot be meaningfully split up into its three constituent parts, other than by making the grossest of assumptions, but if we go down to the level of individual branches of production and services, much more reliable estimates can be made, because clearly not all branches contained all three labour relations. For all government services, for example, there are no employers, and, in most cases, neither can there be expected to be much self-employment here. Other sectors, to the contrary, can be assumed to have been largely based on self-employment, based on what we know from the literature. This way, combining methods of deduction and elimination we can arrive at informed branch-by-branch estimates.

This procedure involves making one central assumption, however, regarding the share of labour relation 13 - "Employers", because without this we are confronted with an insoluble puzzle in the majority of cases. No other data-sets exist which can provide us with a reliable estimate of the number of employers in late nineteenth-early twentieth century Russia, and therefore we have resorted to data-sets from the Collaboratory on other countries to come up with a likely guesstimate for the average percentage of employers in societies at a comparable stage of development. The available data-sets for the benchmark 1900 contain the following percentages (of all persons employed in commodified labour):

Table 1 – Employers as a percentage of commodified labour (ca 1900)

	Spain[68]	Italy[69]	Netherlands[70]
Men	6.2%	6.4%	-
Women	0.7%	5.3%	-
Without subdivision by sex	5.2%	6.1%	4.4%

[66] Troinitskii, N.A. (ed), *Pervaia vseobshchaia perepis' naseleniia Rossiiskoi imperii 1897 g. Chislennost' i sostav rabochikh v Rossii na osnovanii dannykh Pervoi vseobshchei perepisi naseleniia Rossiiskoi imperii 1897 g.* (St. Petersburg, 1906), pp. III, Vol. 1.

[67] Troinitskii, N.A. (ed), *Pervaia vseobshchaia perepis' naseleniia Rossiiskoi imperii 1897 g. Chislennost' i sostav rabochikh v Rossii na osnovanii dannykh Pervoi vseobshchei perepisi naseleniia Rossiiskoi imperii 1897 g.*

[68] Lana-Berasain, José-Miguel, 2016, "Spain 1800,1900,2000 [Global Collaboratory on the History of Labour Relations 1500-2000 Dataset]", http://hdl.handle.net/10622/CH6ZP5, IISH Dataverse, V1

[69] Caracausi, Andrea, 2016, "Italy 1900,2000 [Global Collaboratory on the History of Labour Relations 1500-2000 Dataset]", http://hdl.handle.net/10622/TKD5EC, IISH Dataverse, V1

[70] Gerwen, Jacques van, 2016, "Netherlands 1900,2000 [Global Collaboratory on the History of Labour Relations 1500-2000 Dataset]", http://hdl.handle.net/10622/SIBVDA, IISH Dataverse, V1

Of these, the Netherlands clearly was not at a comparable stage of development with Russia around 1900, and therefore we disregard this figure. Leaves us with Spain and Italy, which can be regarded as being more or less in Russia's peer-group at that point in time. Although the number of observations is clearly not large enough to draw solid conclusions, the percentage of employers among men is similar in Spain and Italy, and on this basis we set the percentage of employers for Russia at an equal 6%.

For women the available data differ too widely, however — in Italy the percentage is slightly lower than for men, but more or less at the same level, whereas for Spain it is much lower. For further guidance we therefore turned to the literature. In her study on female entrepreneurs Galina Ulianova has lined up some of the quantitative evidence on female employers in the 1897 Russian Empire. In industry 4.8% of enterprises was owned by women, and these factories employed 5.6% of the total workforce in industry. The 1711 female entrepreneurs who owned these enterprises accounted for 0.095% of the total workforce in industry.[71] This is a figure which can be compared to a more or less similar figure which can be calculated for 1900 Spain, where 8839 female employers accounted for 0.129% of the workforce in the economy as a whole. These percentages are in the same range, and this has led us to adopt the Spanish percentage of 0.7% female employers as expressed as a percentage of commodified labour (cf. table 1) for 1900 Russia as well.

Next step is to combine the estimates of the share of employers with the data on manual workers on a branch-by-branch basis and make qualified assumptions on the extent to which the residual category contains only self-employed, white-collar workers, or both. In the first two cases, the puzzle is solved, whereas in the last case additional assumptions or calculations are required to separate the two.

Because the data on manual labourers are available only at the highest level of aggregation in the two-tier occupational classification adhered to in the 1897 census, this exercise can only be carried out at this aggregate level, and the detailed breakdown into occupational groups present in the April2019 data-set has to be sacrificed in the Sept2019 data-set. We did use some of the information from this sub-sectoral breakdown in the split-up procedure, though, because in certain cases it allowed us to arrive at better estimates.

The table below renders the followed procedure and assumptions in detail:

Table 2 – Split-up of commodified labour relations by branch, Russia, 1897

Branch	Blue-collar workers[72]	Employers	White-collar workers or other employed[73]	Self-employed	Comments
01-Administration, courts and police	NO	NO	YES	NO	
02-Public (rural, urban) and estates service	NO	NO	YES	NO	
03-Private legal services	NO	For women assumed to be zero, for men estimated as 6% of Sub-category: "03.01-Notaries, jurors, private legal agents"	Sub-category: "03.02-Workers at offices of notaries and legal agents; attendants and messengers"	For women assumed to be zero, for men: Sub-category: "03.01-Notaries, jurors, private legal agents", minus employers.	Assumes that the few women who were employed in this branch, will have worked in employment, rather than being employers or self-employed

[71] Ul'ianova, G.N. "Zhenshchiny-predprinimateli Rossiiskoi imperii v 1890-e gg.: ekonomicheskaia deiatel'nost', sotsiobiograficheskie i etnicheskie parametry", in: *Ekonomicheskaia istoriia. Ezhegodnik 2016/17* (Moskva, 2017), pp. 140-169.

[72] Note, that this is according to the definition as adhered to in the census, which lists the number of manual workers (Russ: *rabochie*) in the various branches.

[73] People working in employment, but not listed as manual workers (Russ: *rabochie*) in the census.

04-Armed forces	NO	NO	YES	NO	Concerns only professional military personnel: lower-ranking conscripted soldiers in army and navy included in labour relation 8.
05-Ministry in the Orthodox faith	NO	NO	YES	NO	
06-Ministry in other Christian faiths	NO	NO	YES	NO	
07-Ministry in non-Christian faiths	NO	NO	YES	NO	
08-Officials at churches, mosques, synagogues, prayer houses, cemetaries, etc., their servants and caretakers	NO	NO	YES	NO	
09-Education services	NO	NO	All, except those in sub-category "09.05-Private teachers (outside of educational institutions)"	Sub-category: "09.05-Private teachers (outside of educational institutions)"	
10-Science, literature, and art	NO	Estimated as 6% of men, and 0.7% of women in sub-category "10.04-White-collar workers at academic and artistic institutions and societies, theaters, museums, newspapers and magazines"	Sub-categories: "10.02-Engineers and technicians" & "10.04-White-collar workers at academic and artistic institutions and societies, theaters, museums, newspapers and magazines " [MINUS employers.] & "10.05-Servants at academic and artistic institutions, theaters", minus employers. PLUS 50% of sub-categories "10.01-Academics and literary figures" & "10.03-Artists, musicians, theater performers and actors"	50% of sub-categories "10.01-Academics and literary figures" & "10.03-Artists, musicians, theater performers and actors" PLUS sub-category "10.06-Unspecified within group"	Sub-categories "10.01-Academics and literary figures" & "10.03-Artists, musicians, theater performers and actors" split up between white-collar workers and self-employed on an equal basis.

11-Medical and health services	NO	Estimated as 6% of men, and 0.7% of women in sub-categories: "11.01-Administrators at healthcare institutions, civilian doctors" & "11.04-Veterinarians" & "11.05-Druggists, pharmacists" & "11.06-Dentists"	All, except those in sub-categories "11.03-Midwives, delivery nurses" & "11.11-Practicing medicine without a medical license (folk healers, peasant midwives, self-taught veterinarians, etc.)" MINUS employers.	Sub-categories: "11.03-Midwives, delivery nurses" & "11.11-Practicing medicine without a medical license (folk healers, peasant midwives, self-taught veterinarians, etc.)"	Ignores the existence of independently practicing doctors and dentists in self-employment, but there is no way of assessing this and the bias is to some degree corrected by the fact that the categories 11.03 and 11.11 will in reality have contained not exclusively self-employed.
12-Service in charitable institutions; 13-Private activity and service; servants, day laborers	Listed in census	NO	All, except those listed as manual workers.	NO	Branches 12 and 13 cannot be separated in data on manual labour, therefore united.
14-Subsisting on income from capital, real estate or funds from parents and relatives	Not applicable: labour relations 1 & 2				
15-Subsisting at expense of public treasury, social institutions and private individuals	Not applicable: labour relation 1				
16-Deprived of freedom and serving sentence	Not applicable: labour relations 1 and 8				
17-Agriculture; 18-Apiculture and sericulture; 19-Livestock farming	Listed in census	NO	NO	NO	Concerns only those listed as manual labourers in agriculture in the census. Peasant producers included in labour relation 4.
20-Forestry, forest industries	Listed in census	Estimated as 6% of the workforce for men, and 0.7% for women.	Non-manual workers in industry and trade all listed in category 13.	Residual category after subtracting manual workers and employers from the total workforce in this branch.	
21-Fishing and hunting	Listed in census	Estimated as 6% of the workforce for men, and 0.7% for women.	Non-manual workers in industry and trade all listed in category 13.	Residual category after subtracting manual workers and employers from the total workforce in this branch.	

22-Mining and quarrying	Listed in census	Estimated as 6% of the workforce for men, and 0.7% for women.	Non-manual workers in industry and trade all listed in category 13.	Residual category after subtracting manual workers and employers from the total workforce in this branch.	
23-Smelting of metals	Listed in census	Estimated as 6% of the workforce for men, and 0.7% for women.	Non-manual workers in industry and trade all listed in category 13.	Residual category after subtracting manual workers and employers from the total workforce in this branch.	
24-Processing of fibrous materials	Listed in census	Estimated as 6% of the workforce for men, and 0.7% for women.	Non-manual workers in industry and trade all listed in category 13.	Residual category after subtracting manual workers and employers from the total workforce in this branch.	
25-Processing of animal products	Listed in census	Estimated as 6% of the workforce for men, and 0.7% for women.	Non-manual workers in industry and trade all listed in category 13.	Residual category after subtracting manual workers and employers from the total workforce in this branch.	
26-Processing of wood	Listed in census	Estimated as 6% of the workforce for men, and 0.7% for women.	Non-manual workers in industry and trade all listed in category 13.	Residual category after subtracting manual workers and employers from the total workforce in this branch.	
27-Processing of metals	Listed in census	Estimated as 7% of the workforce for men, and 0.7% for women.	Non-manual workers in industry and trade all listed in category 13.	Residual category after subtracting manual workers and employers from the total workforce in this branch.	
28-Processing of mineral substances (ceramics)	Listed in census	Estimated as 6% of the workforce for men, and 0.7% for women.	Non-manual workers in industry and trade all listed in category 13.	Residual category after subtracting manual workers and employers from the total workforce in this branch.	

29-Production of chemicals and related production	Listed in census	Estimated as 6% of the workforce for men, and 0.7% for women.	Non-manual workers in industry and trade all listed in category 13.	Residual category after subtracting manual workers and employers from the total workforce in this branch.	
30-Distilling, brewing and honey fermenting	Listed in census	Estimated as 6% of the workforce for men, and 0.7% for women.	Non-manual workers in industry and trade all listed in category 13.	Residual category after subtracting manual workers and employers from the total workforce in this branch.	
31-Other beverages and fermented substances	Listed in census	Estimated as 6% of the workforce for men, and 0.7% for women.	Non-manual workers in industry and trade all listed in category 13.	Residual category after subtracting manual workers and employers from the total workforce in this branch.	
32-Processing of plant and animal food products	Listed in census	Estimated as 6% of the workforce for men, and 0.7% for women.	Non-manual workers in industry and trade all listed in category 13.	Residual category after subtracting manual workers and employers from the total workforce in this branch.	
33-Tobacco and tobacco products	Listed in census	Estimated as 6% of the workforce for men, and 0.7% for women.	Non-manual workers in industry and trade all listed in category 13.	Residual category after subtracting manual workers and employers from the total workforce in this branch.	
34-Printing	Listed in census	Estimated as 6% of the workforce for men, and 0.7% for women.	Non-manual workers in industry and trade all listed in category 13.	Residual category after subtracting manual workers and employers from the total workforce in this branch.	
35-Manufacture of instruments (physical, optical, surgical), watches, toys	Listed in census	Estimated as 6% of the workforce for men, and 0.7% for women.	Non-manual workers in industry and trade all listed in category 13.	Residual category after subtracting manual workers and employers from the total workforce in this branch.	

36-Jewelry making, painting, manufacture of cultural, luxury items, etc.	Listed in census	Estimated as 6% of the workforce for men, and 0.7% for women.	Non-manual workers in industry and trade all listed in category 13.	Residual category after subtracting manual workers and employers from the total workforce in this branch.	
37-Garment making	Listed in census	Estimated as 6% of the workforce for men, and 0.7% for women.	Non-manual workers in industry and trade all listed in category 13.	Residual category after subtracting manual workers and employers from the total workforce in this branch.	
38-Construction, repair and maintenance of residences and general construction work	Listed in census	Estimated as 6% of the workforce for men, and 0.7% for women.	Non-manual workers in industry and trade all listed in category 13.	Residual category after subtracting manual workers and employers from the total workforce in this branch.	
39-Manufacture of carriages and construction of wooden vessels	Listed in census	Estimated as 6% of the workforce for men, and 0.7% for women.	Non-manual workers in industry and trade all listed in category 13.	Residual category after subtracting manual workers and employers from the total workforce in this branch.	
40-Manufacture not included in above or undetermined	Listed in census	Estimated as 6% of the workforce for men, and 0.7% for women.	Residual category after subtracting manual workers and employers from the total workforce in this branch.	Description of sub-groups does not suggest the presence of self-employment in this category.	
41-Water transport	Listed in census	Estimated as 6% of the workforce for men, and 0.7% for women.	Residual category after subtracting manual workers and employers from the total workforce in this branch.	Description of sub-groups does not suggest the presence of substantial self-employment in this category.	
42-Railroads	Listed in census	NO	All, except those listed as manual workers.	NO	
43-Carrier trade	Listed in census	Estimated as 6% of the workforce for men, and 0.7% for women.	NO	All, except those listed as manual workers MINUS employers.	Branch of industry dominated by self-employed peasant migrant labourers (*otkhodniki*).[74]
44-Other overland transport and means of transport	Listed in census	Estimated as 6% of the workforce for men, and 0.7% for women.	Residual category after subtracting manual workers and employers from the total workforce in this branch.	Description of sub-groups does not suggest the presence of substantial self-employment in this category.	

[74] Bradley, J., *Muzhik and Muscovite. Urbanization in Late Imperial Russia* (Berkeley, 1985), pp. 155-182.

45-Mail, telegraph, and telephone	Listed in census	Estimated as 6% of the workforce for men, and 0.7% for women.	Residual category after subtracting manual workers and employers from the total workforce in this branch.	Description of sub-groups does not suggest the presence of self-employment in this category.	
46-Credit and public commercial institutions	NO	Estimated as 6% of the workforce for men, and 0.7% for women.	All, minus employers.	Description of sub-groups does not suggest the presence of self-employment in this category.	
47-Commercial brokerage services; 48-Trade overall; 49-Trade in live livestock; 50-Trade in grain products; 51-Trade in other agricultural products; 52-Trade in construction materials and fuels; 53-Trade in household articles; 54-Trade in metal goods, machines and weapons; 55-Trade in fabrics and garments; 56-Trade in leather, fur, etc.; 57-Trade in articles of luxury, science, art, culture, etc.; 58-Trade in other articles; 59-Peddling, street hawking	Listed in census for all branches of trade taken together, except 61 - Drinking establishments	Estimated as 6% of the workforce for men, and 0.7% for women.	Non-manual workers in industry and trade all included in category 13.	Residual category after subtracting employers and manual workers from the total workforce in these branches.	
60-Inns, hotels, rooming houses and clubs	Listed in census	Estimated as 6% of the workforce for men, and 0.7% for women.	All, except manual workers and employers, split equally between self-employment and white-collar workers.		No sensible criteria available to come to a more meaningful split-up of the residual category of white-collar workers and self-employment.
61-Drinking establishments	Listed in census	Estimated as 6% of the workforce for men, and 0.7% for women.	All, except manual workers and employers, split equally between self-employment and white-collar workers.		No sensible criteria available to come to a more meaningful split-up of the residual category of white-collar workers and self-employment.

62-Establishments providing bathing and personal hygiene services	Listed in census	Estimated as 6% of the workforce for men, and 0.7% for women.	All, except manual workers and employers, split equally between self-employment and white-collar workers.		No sensible criteria available to come to a more meaningful split-up of the residual category of white-collar workers and self-employment.
63-Persons of undetermined occupation	NO	NO	NO	Assumed that all those of undetermined occupation were self-employed.	
64-Prostitution	NO	Sub-category: "64.01-Operation of brothels"	Sub-category: "64.02-Prostitutes"	NO	
65-Unknown occupations	Cannot be meaningfully split up, remains as combined 12ab131418				

Next and final step is to assign taxonomy codes to these four categories. This is entirely straightforward for the self-employed and employers, who translate easily into 12a/b - "Self-employed leading producers" and 13 -"Employers", but the distinction between manual workers and white-collar workers is not something that coincides with the taxonomy, which distinguishes instead between wage and salary workers working for the market (LabRel 14) and those working in non-market institutions (LabRel 18). For certain sectors we can make qualified assumptions, though, and where possible we have distinguished between Labour relation 14 and Labour relation 18, but in most cases we have no choice but settling for a combined Labour relation 1418 - "Wage earners in market and non-market institutions".

Applying these principles and redistribution keys to the figures for the combined category 12a/b131418 we produced a second version of the 1900_Russia dataset, labeled Sept2019, in which the data for commodified labour are split up between the labour relations 12a/b, 13, 14, 18 and 1418, with a small residual category 12a/b131418, about which we have no further information and which can therefore not be split up. As explained above, the split-up comes at the expense of the detailed breakdown into occupational categories which is present in the April2019 data-set.

Table 3 provides an overview of how the two versions of the data-set compare. For purposes of comparison the (incomplete) separate figures for labour relations 14 and 18 have been folded into the over-arching category 12a/12b/13/14/18 (April data-set) and 14/18 (September data-set):

Table 3 – Labour relations Russia - 1900

Labour relation	April 2019		Sept 2019		Labour relation
	Absolute numbers	Percentages	Absolute numbers	Percentages	
1 - "Cannot work or cannot be expected to work"	21,997,887	22.2	21,997,887	22.2	1 - "Cannot work or cannot be expected to work"
2 - "Affluent"	437,749	0.4	437,749	0.4	2 - "Affluent"
4a - "Leading household producers"	12,162,729	12.3	12,162,729	12.3	4a - "Leading household producers"
4b/5 - "Kin producers and Kin non-producers"	51,861,799	52.3	51,861,799	52.3	4b/5 - "Kin producers and Kin non-producers"
8 - "Obligatory labourers"	694,286	0.7	694,286	0.7	8 - "Obligatory labourers"
12a/12b/13/14/18 - "Commodified labour"	12,047,237	12.1	3,210,697	3.2	12a/12b - "Self-employed"
			311,852	0.3	13 - "Employers"
			8,360,885	8.4	14/18 - "Wage earners"
			163,803	0.2	12a/12b/13/14/18 - "Commodified labour"
TOTAL	99,201,687	100	99,201,687	100	TOTAL

The table above provides the aggregate data for the territory of European Russia and Siberia, which roughly corresponds to the territory of the 1800 Russian Empire. In the data-set separate figures are available for European Russia and Siberia, as well as for men and women.

Appendix IV – Look-up table census-categories 1900 & 2000 (EN<>RUS)

Cross-section 1900

ENGLISH		RUSSIAN
01-Administration, courts and police		01-Администрация, суд и полиция
	01.01-Ranks in administrative and judicial departments	01.01-Чины административного и судебного ведомства
	01.02-Ranks in land survey department	01.02-Чины межевого ведомства
	01.03-Workers in offices of administrative and judicial departments	01.03-Служащие в канцеляриях административного и судебного ведомства
	01.04-Attendants, guards, etc. at administrative and judicial offices	01.04-Служители, сторожа и т.п. административного и судебного ведомства
	01.05-Court ranks, white-collar workers at palaces in general	01.05-Придворные чины и вообще служащие при дворцах
	01.06-Foreign ambassadors and white-collar workers at embassies and missions	01.06-Иностранные послы и служащие в посольствах и миссиях
	01.07-Officer ranks of gendarmerie and police	01.07-Офицерские чины жандармов и полиции
	01.08-Lower ranks of gendarmerie and police (urban and rural), firemen	01.08-Нижние чины жандармов и полиции городской и сельской, пожарные
	01.09-Unspecified within group	01.09-Нераспределены внутри группы
02-Public (rural, urban) and estates service		02-Общественная (земская, городская) и сословная служба
	02.01-Elected civil servants at public and estates institutions	02.01-Служащие по выборам в общественных и сословных учреждениях
	02.02-Stanitsa atamans, gmina prefects, volost and village leaders, etc.	02.02-Станичные атаманы, гминные войты, волостные и сельские старшины и т.п.
	02.03-Hired civil servants at public and estates institutions	02.03-Служащие по найму в общественных и сословных учреждениях
	02.04-Attendants, messengers, guards, etc. at public and estates institutions	02.04-Служители, рассыльные, сторожа и т.п. в общественных и сословных учреждениях
	02.05-Unspecified within group	02.05-Нераспределены внутри группы
03-Private legal services		03-Частная юридическая деятельность
	03.01-Notaries, jurors, private legal agents	03.01-Нотариусы, присяжные и частные поверенные
	03.02-Workers at offices of notaries and legal agents; attendants and messengers	03.02-Занимающиеся в конторах нотариусов и поверенных. Служители и рассыльные
	03.03-Unspecified within group	03.03-Нераспределены внутри группы
04-Armed forces		04-Вооруженные силы
	04.01-Generals, staff and company officers of military department	04.01-Генералы, штаб- и обер-офицеры военного ведомства
	04.02-Lower ranks	04.02-Нижние чины
	04.03-Military topographical engineers	04.03-Военные топографы
	04.04-Civilian ranks in military department	04.04-Гражданские чины военного ведомства
	04.05-Workers in the offices of military department	04.05-Служащие в канцеляриях военного ведомства
	04.06-Attendants and guards at military department	04.06-Служители и сторожа военного ведомства
	04.07-Admirals, staff and company officers of fleet	04.07-Адмиралы, штаб- и обер-офицеры флота
	04.08-Lower ranks of fleet	04.08-Нижние чины флота
	04.09-Civilian ranks of marine department	04.09-Гражданские чины морского ведомства
	04.10-Workers in the offices of marine department	04.10-Служащие в канцеляриях морского ведомства
	04.11-Attendants and guards at marine department	04.11-Служители и сторожа морского ведомства
	04.12-Generals, staff and company officers of border guard	04.12-Генералы, штаб- и обер-офицеры пограничной стражи
	04.13-Lower ranks of border guard	04.13-Нижние чины пограничной стражи

	04.14-Civilian ranks of border guard corps	04.14-Гражданские чины корпуса пограничной стражи
	04.15-Workers in the offices of border guard corps	04.15-Служащие в канцеляриях корпуса пограничной стражи
	04.16-Attendants and guards in border guard corps	04.16-Служители и сторожа корпуса пограничной стражи
	04.17-Unspecified within group	04.17-Нераспределены внутри группы
05-Ministry in the Orthodox faith	05-Богослужение православного исповедания	
	05.01-Monks and nuns, Orthodox	05.01-Монашествующие православного исповедания
	05.02-Novices	05.02-Послушники и послушницы
	05.03-Clergy, Orthodox	05.03-Священнослужители православного исповедания
	05.04-Acolytes, parish clerks, church singers, communion bread makers	05.04-Церковно-служители, псаломщики, церковные певчие, просфорницы и т.д.
	05.05-Readers for the dead, collectors of church donations	05.05-Чтецы и чтицы по усопшим, сборщики подаяний на церкви
	05.06-Unspecified within group	05.06-Нераспределены внутри группы
06-Ministry in other Christian faiths	06-Богослужение других христианских исповеданий	
	06.01-Monks and nuns, other Christian faiths	06.01-Монашествующие других христианских исповеданий
	06.02-Clergy, other Christian faiths	06.02-Священнослужители других христианских исповеданий
	06.03-Acolytes, other Christian faiths	06.03-Церковнослужители других христианских исповеданий
	06.04-Unspecified within group	06.04-Нераспределены внутри группы
07-Ministry in non-Christian faiths	07-Богослужение нехристианских исповеданий	
	07.01-Major orders, non-Christian faiths	07.01-Высшие духовные лица нехристианских исповеданий
	07.02-Minor orders, non-Christian faiths	07.02-Низшие духовные лица нехристианских исповеданий
	07.03-Unspecified within group	07.03-Нераспределены внутри группы
08-Officials at churches, mosques, synagogues, prayer houses, cemetaries, etc., their servants and caretakers	08-Должностные лица при церквах, молельнях, мечетях, синагогах, кладбищах и т.п., прислуга и сторожа при них	
	08.01-Chapel keepers, adherents of dissenting doctrine, members of various religious communities irrespective of faith	08.01-Содержатели молелен, раскольничьи начетчики, члены разных религиозных обществ без различия исповеданий
	08.02-Attendants and guards at churches, monasteries, temples and cemeteries	08.02-Служители и сторожа при церквах и монастырях, храмах и кладбищах
	08.03-Unspecified within group	08.03-Нераспределены внутри группы
09-Education services	09-Учебная и воспитательная деятельность	
	09.01-Administrators and teachers at educational institutions	09.01-Начальники и учащие в учебных заведениях
	09.02-Other ranks (besides administrators and teachers) at educational institutions	09.02-Прочие (кроме начальников и учащих) чины учебных заведений
	09.03-Workers in offices of educational institutions	09.03-Служащие в канцеляриях учебных заведений
	09.04-Guards and servants at educational institutions	09.04-Сторожа и прислуга в учебных заведениях
	09.05-Private teachers (outside of educational institutions)	09.05-Частные (вне учебных заведений) преподаватели и преподавательницы
	09.06-Tutors and governesses employed by private individuals	09.06-Гувернеры и гувернантки, служащие у частных лиц
	09.07-Teachers of arts and crafts	09.07-Учители искусств и ремесел
	09.08-Unspecified within group	09.08-Нераспределены внутри группы
10-Science, literature, and art	10-Наука, литература и искусства	
	10.01-Academics and literary figures	10.01-Ученые и литераторы
	10.02-Engineers and technicians	10.02-Инженеры и технологи

	10.03-Artists, musicians, theater performers and actors	10.03-Художники, музыканты, театральные артисты и актеры
	10.04-White-collar workers at academic and artistic institutions and societies, theaters, museums, newspapers and magazines	10.04-Служащие в ученых и художественных учреждениях и обществах, театрах, музеях, в редакциях газет и журналов
	10.05-Servants at academic and artistic institutions, theaters	10.05-Прислуга в учреждениях ученых и художественных и в театрах
	10.06-Unspecified within group	10.06-Нераспределены внутри группы
11-Medical and health services		11-Врачебная и санитарная деятельность
	11.01-Administrators at healthcare institutions, civilian doctors	11.01-Начальники лечебных заведений, гражданские врачи
	11.02-Army and navy doctors	11.02-Врачи армии и флота
	11.03-Midwives, delivery nurses	11.03-Акушерки и повивальные бабки
	11.04-Veterinarians	11.04-Ветеринары
	11.05-Druggists, pharmacists	11.05-Аптекари, провизоры, фармацевты
	11.06-Dentists	11.06-Дантисты
	11.07-Medics, pharmacy trainees, etc.	11.07-Фельдшера, аптекарские ученики и т.п.
	11.08-Nurses, supervisors in hospitals	11.08-Сестры милосердия, надзиратели и надзирательницы в больницах
	11.09-Attendants and servants in hospitals and pharmacies	11.09-Сиделки и прислуга в больницах и аптеках
	11.10-Orderlies, disinfectors	11.10-Санитары и дезинфекторы
	11.11-Practicing medicine without a medical license (folk healers, peasant midwives, self-taught veterinarians, etc.)	11.11-Занимающиеся врачеванием без права на врачебную деятельность (знахари, повитухи, коновалы и т.п.)
	11.12-Unspecified within group	11.12-Нераспределены внутри группы
12-Service in charitable institutions		12-Служба при благотворительных учреждениях
	12.01-Workers at shelters, almshouses, orphanages and other charitable institutions	12.01-Служащие в приютах, богадельнях, воспитательных домах и др. благотворительных учреждениях
13-Private activity and service; servants, day laborers		13-Деятельность и служба частная. Прислуга, поденщики
	13.01-White-collar workers and servants at trade and industrial enterprises	13.01-Служащие и прислуга в торговых и промышленных предприятиях
	13.02-Administration in trade and industrial enterprises	13.02-Администрация в торговых и промышленных предприятиях
	13.03-Employees in management of railway and steamship companies	13.03-Служащие в правлениях железнодорожных и пароходных обществ
	13.04-White-collar workers at private institutions	13.04-Служащие в частных учреждениях
	13.05-Servants at private institutions	13.05-Прислуга в частных учреждениях
	13.06-White-collar employees of private individuals: managers, secretaries, stewards, etc.	13.06-Служащие у частных лиц: управляющие, секретари, старосты и т.п.
	13.07-Domestic employees: housekeepers, chaperons, lectresses, nursery governesses, etc.	13.07-Домашние служащие: экономки, компаньонки, лектрисы, бонны и т.п.
	13.08-Servants at estates, factories and plants	13.08-Прислуга в усадьбах, на фабриках и заводах
	13.09-House servants: doormen, caretakers, nightwatchmen, etc.	13.09-Прислуга домовая: швейцары, дворники, ночные сторожа и т.п.
	13.10-Domestic servants: cooks, footmen, maids, etc.	13.10-Прислуга домашняя: повара, лакеи, кухарки, горничные и т.п.
	13.11-Day and menial laborers without specific specialization	13.11-Поденщики и чернорабочие без определенной специальности
	13.12-Unspecified within group	13.12-Нераспределены внутри группы
14-Subsisting on income from capital, real estate or funds from parents and relatives		14-Живущие доходом с капитала и недвижимого имущества или на средства, получаемые от родителей и родственников
	14.01-Subsisting on income from capital and real estate	14.01-Живущие доходами с капитала и недвижимого имущества
	14.02-Subsisting on funds from parents or other relatives	14.02-Живущие на средства, получаемые от родителей или родственников

	14.03-Subsisting on pension	14.03-Живущие пенсией
	14.04-Unspecified within group	14.04-Нераспределены внутри группы
15-Subsisting at expense of public treasury, social institutions and private individuals		15-Живущие на счет казны и общественных учреждений и на средства частных лиц
	15.01-Pupils at boarding schools, private boarding houses, etc.	15.01-Воспитанники закрытых учебных заведений, частных пансионов и т.п.
	15.02-Indigents at almshouses, shelters, etc.	15.02-Призреваемые в богадельнях, приютах и т.п.
	15.03-Children at orphanages	15.03-Питомцы воспитательных домов
	15.04-Patients in hospitals	15.04-Больные в больницах
	15.05-Stipend recipients of various institutions and private individuals	15.05-Стипендиаты различных учреждений и частных лиц
	15.06-Subsisting on assistance at place of residence	15.06-Живущие пособием на местах жительства
	15.07-Unspecified within group	15.07-Нераспределены внутри группы
16-Deprived of freedom and serving sentence		16-Лишенные свободы и отбывающие наказание
	16.01-Convicts in prisons	16.01-Арестанты в тюрьмах
	16.02-Exiles, penal colony convicts receiving allowance from the state	16.02-Ссыльные и ссыльно-каторжные, получающие довольствие от казны
	16.03-Unspecified within group	16.03-Нераспределены внутри группы
17_18_19-Agriculture; Apiculture & sericulture; Livestock farming		17_18_19-Сельское хозяйство; пчеловодство и шелководство; животноводство
	17_18_19-(a)Not working for hire in agriculture; apiculture; sericulture; livestock farming	17_18_19-(а)Работающие не по найму в сельском хозяйстве, пчеловодстве, шелководстве и животноводстве
	17_18_19-(b)Working for hire in agriculture; apiculture; sericulture; livestock farming	17_18_19-(б)Работающие по найму в сельском хозяйстве, пчеловодстве, шелководстве и животноводстве
17-Agriculture		17-Сельское хозяйство
	17.01-Crop farming overall	17.01-Земледелие вообще
	17.02-Farming of flax, hemp and other fiber plants	17.02-Льноводство, разведение конопли и др. волокнистых растений
	17.03-Rice farming	17.03-Культура риса
	17.04-Farming of melons, gourds, beets and other specialty crops	17.04-Бахчеводство, свекловодство и др. специальные культуры
	17.05-Market gardening	17.05-Огородничество
	17.06-Cotton farming	17.06-Хлопководство
	17.07-Tobacco farming	17.07-Табаководство
	17.08-Tea cultivation	17.08-Культура чая
	17.09-Vegetable, fruit and berry farming	17.09-Садоводство и плодоводство
	17.10-Winemaking	17.10-Виноделие
	17.11-Unspecified within group	17.11-Нераспределены внутри группы
18-Apiculture and sericulture		18-Пчеловодство и шелководство
	18.01-Apiculture	18.01-Пчеловодство
	18.02-Sericulture	18.02-Шелководство
	18.03-Unspecified within group	18.03-Нераспределены внутри группы
19-Livestock farming		19-Животноводство
	19.01-Livestock farming overall	19.01-Скотоводство вообще
	19.02-Horse breeding, selective	19.02-Коневодство заводское
	19.03-Horse breeding, pasture	19.03-Коневодство косячное
	19.04-Other sectors of livestock farming	19.04-Прочие отрасли животноводства
	19.05-Unspecified within group	19.05-Нераспределены внутри группы
20-Forestry, forest industries		20-Лесоводство и лесные промыслы
	20.01-Forestry	20.01-Лесоводство
	20.02-Forest industries, harvesting of forest materials, chopping and sawing firewood, bast production	20.02-Лесные промыслы, заготовка лесных материалов, рубка и пилка дров, изготовление луба и т.п.
	20.03-Preparation of forest materials for factories and mines	20.03-Заготовка лесных материалов для заводов и рудников
	20.04-Production of charcoal	20.04-Производство древесного угля

	20.05-Cottage forest industries (distilling wood tar and resins, calcination of charcoal)	20.05-Кустарные лесные промыслы (сидка дегтя и смолы, жжение угля)
	20.06-Secondary forest industries (gathering berries, mushrooms, etc.)	20.06-Побочные лесные промыслы (сбор ягод, грибов и т.п.)
	20.07-Unspecified within group	20.07-Нераспределены внутри группы
21-Fishing and hunting	21-Рыболовство и охота	
	21.01-Fishing, saltwater	21.01-Рыболовство морское
	21.02-Fishing, freshwater	21.02-Рыболовство речное
	21.03-Marine mammal hunting	21.03-Морские звериные промыслы
	21.04-Specialized saltwater and freshwater industries: pearl diving, harvesting seaweed, leech trapping, etc.	21.04-Особые морские и речные промыслы: ловля жемчуга, добыча морской капусты, ловля пиявок и т.п.
	21.05-Animal trapping and commercial hunting	21.05-Звероловство и промысловая охота
	21.06-Unspecified within group	21.06-Нераспределены внутри группы
22-Mining and quarrying	22-Добыча руд и копи	
	22.01-Extractive industries overall	22.01-Горная промышленность вообще
	22.02-Mining of hard and brown coal, anthracite, etc.	22.02-Добыча каменного и бурого угля, антрацита и т.п.
	22.03-Harvesting of peat	22.03-Добыча торфа
	22.04-Extraction of oil	22.04-Добыча нефти
	22.05-Mining of iron ore	22.05-Добыча железных руд
	22.06-Mining of copper ore	22.06-Добыча медных руд
	22.07-Mining of silver and lead ore	22.07-Добыча серебряно-свинцовых руд
	22.08-Mining of manganese ore	22.08-Добыча марганцевых руд
	22.09-Mining of zinc ore	22.09-Добыча цинковых руд
	22.10-Mining of cinnabar	22.10-Добыча ртутных руд
	22.11-Mining of chrome, cobalt, nickel and other ores	22.11-Добыча хромовых, кобальтовых, никелевых и прочих руд
	22.12-Mining of gold and platinum	22.12-Добыча золота и платины
	22.13-Mining of table and other salt	22.13-Добыча поваренной и прочих солей
	22.14-Mining of semi-precious and precious stones	22.14-Добыча цветных и драгоценных камней
	22.15-Quarrying and mining of natural construction materials	22.15-Каменоломни и добыча естественных строительных материалов
	22.16-Production of lime and alabaster	22.16-Производство извести и алебастра
	22.17-Development of mineral springs	22.17-Эксплуатация минеральных источников
	22.18-Unspecified within group	22.18-Нераспределены внутри группы
23-Smelting of metals	23-Выплавка металлов	
	23.01-Cast iron and steel making, iron production	23.01-Выплавка чугуна, стали и выделка железа
	23.02-Smelting of copper, lead, silver, zinc and other metals	23.02-Выплавка меди, свинца, серебра, цинка и др. металлов
	23.03-Unspecified within group	23.03-Нераспределены внутри группы
24-Processing of fibrous materials	24-Обработка волокнистых веществ	
	24.01-Processing of cotton and machine manufacture of cotton products	24.01-Обработка хлопка и фабричное производство изделий из него
	24.02-Dyeing, printing and finishing	24.02-Красильно-набивное и отделочное производство
	24.03-Cottage cotton manufacture	24.03-Кустарное хлопчатобумажное производство
	24.04-Processing of linen, hemp and jute and machine manufacture of products from them	24.04-Обработка льна, пеньки, джута и фабричное производство изделий из них
	24.05-Cottage manufacture of linen and hemp products	24.05-Кустарное производство льняных и пеньковых изделий
	24.06-Manufacture of straw products	24.06-Выделка изделий из соломы
	24.07-Cable and rope industry	24.07-Канатное и веревочное производство

	24.08-Processing of wool and manufacture of wool products	24.08-Обработка шерсти и выделка изделий из нее
	24.09-Cottage manufacture of wool products	24.09-Кустарное производство шерстяных изделий
	24.10-Processing of silk and manufacture of silk fabrics	24.10-Обработка шелка и производство шелковых тканей
	24.11-Cottage silk manufacture, cottage manufacture of brocade, silk brocade, braiding and cannetille	24.11-Кустарно-шелковое производство, кустарная выделка парчи, глазета, галунов и канители
	24.12-Machine manufacture of lace, ruche and tulle	24.12-Кружевное, рюшевое и тюлевое фабричное производство
	24.13-Cottage lace, crochet and embroidery manufacture	24.13-Кустарное кружевное, тамбурное и вышивальное производство
	24.14-Manufacture of hosiery and knitwear	24.14-Производство чулочное и трикотажное
	24.15-Gold and silver embroidery	24.15-Золото- и сереброшвейное производство
	24.16-Braid and trimming products	24.16-Басонные и позументные изделия
	24.17-Felt making	24.17-Войлочное производство
	24.18-Carpet industry	24.18-Ковровое производство
	24.19-Cottage dyeing industry	24.19-Кустарное красильное производство
	24.20-Other sectors of fibrous material processing	24.20-Прочие отрасли обработки волокнистых веществ
	24.21-Unspecified within group	24.21-Нераспределены внутри группы
25-Processing of animal products		25-Обработка животных продуктов
	25.01-Leather tanning	25.01-Выделка кож
	25.02-Sheepskin and fur tanning	25.02-Выделка овчин и мехов
	25.03-Manufacture of leather goods	25.03-Выделка изделий из кож
	25.04-Saddlery	25.04-Шорное и седельное производство
	25.05-Combmaking and manufacture of goods from horns, hooves and bones	25.05-Гребеночное производство и изделия из рога, копыт и кости
	25.06-Sorting and processing of hair, bristle, down and feathers	25.06-Сортировка и обработка волоса, щетины, пуха и пера
	25.07-Manufacture of scrub brushes, paintbrushes, products from bristle and hair, etc.	25.07-Производство щеток, кистей, изделий из щетины и волоса и т.п.
	25.08-Intestine and string industry	25.08-Кишечно-струнное производство
	25.09-Manufacture of glue and gelatin	25.09-Производство клея и желатинное
	25.10-Rendering (raw)	25.10-Живодерное (сырейное) производство
	25.11-Processing of animal products	25.11-Разные производства по обработке животных продуктов
	25.12-Unspecified within group	25.12-Нераспределены внутри группы
26-Processing of wood		26-Обработка дерева
	26.01-Lumber and plywood industry	26.01-Лесопильное и фанерно-пильное производство
	26.02-Bast matting and bast fiber industry	26.02-Рогожное и мочальное производство
	26.03-Manufacture of wheel bands, shaft bows, shafts, staves, small turned and carved goods in general, carriage bodies, runners and sleds	26.03-Производство колесных ободьев, дуг, оглобель, бондарных клепок, щепного товара вообще, экипажей кузовых, ходов и саней
	26.04-Coopering	26.04-Бондарное производство
	26.05-Manufacture of baskets and other wickerwork made from twigs, cane, bast, splint and reed	26.05-Корзиночное производство и иных плетеных изделий из прутьев, луба, лыка и камыша
	26.06-Manufacture of chests, boxes, coffins, etc.	26.06-Сундучное, ящичное, гробовое и т.п. производство
	26.07-Joinery, parquetry, furniture making	26.07-Столярное, паркетное, мебельное производства
	26.08-Manufacture of boot trees, gunstocks, etc.	26.08-Производство сапожных колодок, ружейных ложи и т.п. предметов
	26.09-Woodturning	26.09-Токарное производство

	26.10-Cork industry	26.10-Пробочное производство
	26.11-Manufacture of boot spikes, match sticks, etc.	26.11-Производство сапожных шпилек, спичечной соломки и т.п. изделий
	26.12-Unspecified within group	26.12-Нераспределены внутри группы
27-Processing of metals	27-Обработка металлов	
	27.01-Processing of section iron	27.01-Переработка сортового железа
	27.02-Cast iron and steel making	27.02-Чугунно- и сталелитейное производство
	27.03-Manufacture of lead and tin sheet, lead pipe; type casting, soldering and galvanizing, tinplating, nickel plating, etc.	27.03-Производство свинцовых и оловянных листов, свинцовых труб, словолитное, паяльно-цинковое, лудильное, никелировочное и т.д.
	27.04-Wire and nail manufacturing	27.04-Проволочное и гвоздильное производство
	27.05-Manufacture of hardware	27.05-Производство скобяного товара
	27.06-Manufacture of needles, pins, studs and other metalware	27.06-Игольное, булавочное, шпилечное производство и др. металлических изделий
	27.07-Manufacture of tinware, lamps	27.07-Производство жестяных изделий, ламповое
	27.08-Manufacture of iron, tinplated and enamel dishes	27.08-Производство железной, луженой и эмалированной посуды
	27.09-Metalworking, metalworking-machining and forging (sewing machines and bicycles)	27.09-Производство слесарное, слесарно-механическое и кузнечное (швейных машин и велосипедов)
	27.10-Copper casting, bell and samovar manufacturing	27.10-Меднолитейное, колокольное и самоварное производство
	27.11-Cottage manufacture of various metalware	27.11-Кустарное производство разных металллических изделий
	27.12-Large-scale shipbuilding (steamships, metal vessels)	27.12-Крупное судостроение (пароходы и металлические суда)
	27.13-Machine, ship and train building	27.13-Машинно-судо-паровозо-вагоностроительное производство
	27.14-Manufacture of farming machinery and equipment	27.14-Производство земледельческих машин и орудий
	27.15-Manufacture of firearms, bladed and edged weapons, cartridges and shells	27.15-Производство огнестрельного и холодного оружия, патронов, гильз
	27.16-Unspecified within group	27.16-Нераспределены внутри группы
28-Processing of mineral substances (ceramics)	28-Обработка минеральных веществ (керамические производства)	
	28.01-Milling of mineral substances	28.01-Размол минеральных веществ
	28.02-Brick and roofing tile industry	28.02-Кирпичное и черепичное производство
	28.03-Pottery	28.03-Горшечное и гончарное производство
	28.04-Porcelain and faience industry	28.04-Производство фаянсовое и фарфоровое
	28.05-Manufacture of sculptures from clay and gypsum	28.05-Производство скульптурных произведений из глины и гипса
	28.06-Cement industry, manufacture of artificial stone	28.06-Производство цементное и искусственных камней
	28.07-Asbestos industry	28.07-Производство асбестовое
	28.08-Manufacture of millstones, grinding stones, etc.	28.08-Производство жерновов, точил и т.п.
	28.09-Manufacture of slate pencils, blackboard, chalk and lead pencils	28.09-Производство грифелей, аспидных досок, мелков и карандашей
	28.10-Manufacture of crystal, mirrors and glassware	28.10-Производство хрустальное и зеркальное и изделия из стекла
	28.11-Processing of various mineral substances	28.11-Обработка разных минеральных веществ
	28.12-Unspecified within group	28.12-Нераспределены внутри группы
29-Production of chemicals and related production	29-Производства химические и производства, связанные с ними	
	29.01-Chemical industry overall, chemical activities	29.01-Химическое производство вообще и химическая деятельность
	29.02-Manufacture of acids and salts	29.02-Производство кислот и солей
	29.03-Manufacture of pharmaceutical products and household chemicals	29.03-Производство аптекарских и москательных товаров

	29.04-Manufacture of medicinal essences and preparations	29.04-Производство лекарственных эссенций и препаратов
	29.05-Manufacture of disinfectants, perfume and cosmetic substances	29.05-Производство дезинфицирующих, парфюмерных и косметических веществ
	29.06-Manufacture of photographic preparations	29.06-Производство фотографических препаратов
	29.07-Manufacture of mineral and vegetable pigments	29.07-Производство минеральных и растительных красок
	29.08-Manufacture of explosives	29.08-Производство взрывчатых веществ
	29.09-Manufacture of matches	29.09-Производство спичек
	29.10-Manufacture of fertilizer meal, poudrette, etc.	29.10-Производство удобрительных туков, пудрета и т.п.
	29.11-Manufacture of oil products	29.11-Производство нефтяных продуктов
	29.12-Manufacture of asphalt products	29.12-Производство асфальтовых изделий
	29.13-Manufacture of illuminating gas, coke	29.13-Производство светильного газа, кокса
	29.14-Manufacture of varnishes, drying oils and various ointments	29.14-Производство лаков, олифы и разных мазей
	29.15-Manufacture of the products of dry distillation of wood, turpentine, tar, etc.	29.15-Производство продуктов сухой перегонки дерева, скипидара, дегтя и т.п.
	29.16-Manufacture of sealing wax, pitch and putty	29.16-Производство сургуча, смолок и замазок
	29.17-Manufacture of vulcanized rubber and rubber goods	29.17-Производство резиновых и каучуковых изделий
	29.18-Manufacture of linoleum, oilcloth	29.18-Производство линолеума, клеенок и т.п.
	29.19-Pulp industry	29.19-Производство бумагомассное
	29.20-Paper making	29.20-Производство бумагоделательное
	29.21-Bone roasting and milling	29.21-Костеобжигательное и костемольное производство
	29.22-Albumin industry	29.22-Производство альбуминное
	29.23-Fat melting and bone cooking	29.23-Салотопенное и костеварное производство
	29.24-Soap making	29.24-Мыловаренное производство
	29.25-Candle grease industry	29.25-Производство свечносальное
	29.26-Manufacture of stearin, stearin candles and glycerine	29.26-Производство стеарина, стеариновых свечей и глицерина
	29.27-Wax refining, manufacture of wax candles	29.27-Производство воскобойное и восковых свечей
	29.28-Unspecified within group	29.28-Нераспределены внутри группы
30-Distilling, brewing and honey fermenting		30-Винокурение, пиво- и медоварение
	30.01-Distilling, pure alcohol and vodka	30.01-Производство винокуренное, спиртоочистительное и водочное
	30.02-Beer brewing and honey fermenting	30.02-Производство пиво- и медоваренное
	30.03-Unspecified within group	30.03-Нераспределены внутри группы
31-Other beverages and fermented substances		31-Производство прочих напитков и бродильных веществ
	31.01-Production of sparkling and mineral water, kvas making	31.01-Производство шипучих и минеральных вод и квасоваренное производство
	31.02-Vinegar industry	31.02-Производство уксусное
	31.03-Malting	31.03-Производство солодовенное
	31.04-Yeast industry	31.04-Производство дрожжевое
	31.05-Production of other nutrients (kumis, kefir, etc.)	31.05-Производство других питательных веществ (кумыс, кефир и т.п.)
	31.06-Unspecified within group	31.06-Нераспределены внутри группы
32-Processing of plant and animal food products		32-Обработка растительных и животных питательных продуктов
	32.01-Grain milling and hulling industry	32.01-Мукомольное и круподерное производство
	32.02-Production of pasta and other products from dough	32.02-Производство макаронное и других продуктов из теста

	32.03-Baking industry	32.03-Производство хлебопекарное, булочное, пряничное и т.д.
	32.04-Production of roasted and ground coffee, chicory and other coffee surrogates	32.04-Производство жженого и молотого кофе, цикория и иных суррогатов кофе
	32.05-Production of various condiments and spices; fruit and vegetable preserves	32.05-Производство разных съедобных приправ и пряностей; консервов из фруктов и овощей
	32.06-Confectionery industry	32.06-Производство кондитерское
	32.07-Starch (dextrin) and molasses industry	32.07-Производство крахмальное, паточное и иных продуктов из крахмала (декстрина)
	32.08-Sugar refining industry	32.08-Производство сахароваренное и сахарорафинадное
	32.09-Oilseed processing industry	32.09-Производство маслобойное (растительных масел)
	32.10-Slaughtering industry	32.10-Скотобойный промысел
	32.11-Production of sausage, canned fish and meat	32.11-Производство колбас, консервов рыбных и мясных
	32.12-Butter and cheese industry	32.12-Производство маслодельное и сыроварное
	32.13-Unspecified within group	32.13-Нераспределены внутри группы
33-Tobacco and tobacco products	33-Табак и изделия из него	
	33.01-Processing of tobacco and manufacture of tobacco products	33.01-Обработка табаку и приготовление изделий из него
34-Printing	34-Полиграфические производства	
	34.01-Typography, lithography, phototyping, etc.	34.01-Производство типографское, литографское, фототипное и т.п.
	34.02-Zincography, xylography, metal engraving, etc.	34.02-Цинкографическое, ксилографическое, граверное по металлам и т.п. производства
	34.03-Photography	34.03-Фотография
	34.04-Stationery and book binding, case making, etc.	34.04-Производство брошюровочное, переплетное, футлярное и т.п.
	34.05-Manufacture of cigarette tubes, powder bags, playing cards and other paper products	34.05-Производство гильз папиросных, картузов, конвертов, игральных карт и др. изделий из бумаги
	34.06-Manufacture of wallpaper, dyed and gilded paper	34.06-Производство обоев, крашеной и золоченой бумаги
	34.07-Unspecified within group	34.07-Нераспределены внутри группы
35-Manufacture of instruments (physical, optical, surgical), watches, toys	35-Производство инструментов физических, оптических, хирургических, музыкальных, часов, игрушек и проч.	
	35.01-Watchmaking	35.01-Производство часовое
	35.02-Manufacture of musical instruments	35.02-Производство музыкальных инструментов
	35.03-Manufacture of toys	35.03-Производство игрушек
	35.04-Electroforming	35.04-Гальванопластическое производство
	35.05-Manufacture of physical, optical and surgical instruments, telephone and telegraph accessories, etc.	35.05-Производство физических, оптических, хирургических инструментов, принадлежностей телефона, телеграфа и т.п.
	35.06-Unspecified within group	35.06-Нераспределены внутри группы
36-Jewelry making, painting, manufacture of cultural, luxury items, etc.	36-Ювелирное дело, живопись, производства предметов культа, роскоши и т.д.	
	36.01-Lapidary industry (stone polishing)	36.01-Производство гранильное (шлифовка камней)
	36.02-Manufacture of gold, silver, melchior and bronze items, and jewelry	36.02-Производство золотых, серебряных, мельхиоровых, бронзовых и ювелирных вещей
	36.03-Coin and medal minting	36.03-Монетное производство и изготовление медалей
	36.04-Wood plating and carving, icon case and frame making	36.04-Золочение по дереву, резьба по дереву, производство киотное и багетное
	36.05-Manufacture of papier-mache products	36.05-Производство изделий из папье-маше
	36.06-Icon painting, painting	36.06-Иконопись, живопись
	36.07-Manufacture of liturgical supplies	36.07-Изготовление изделий для богослужебных потребностей
	36.08-Unspecified within group	36.08-Нераспределены внутри группы

37-Garment making		37-Изготовление одежды
	37.01-Manufacture of men's garments	37.01-Изготовление предметов мужской одежды
	37.02-Manufacture of women's dresses	37.02-Изготовление дамских платьев
	37.03-Hatmaking	37.03-Шляпочное, фуражечное и картузо-шапочное производство
	37.04-Manufacture of fashion items	37.04-Изготовление предметов моды
	37.05-Manufacture of gloves, mittens, belts, etc.	37.05-Изготовление перчаток, рукавиц, поясов и проч.
	37.06-Manufacture of footwear	37.06-Изготовление обуви
	37.07-Manufacture of underwear	37.07-Изготовление белья
	37.08-Fabric whitening and dyeing	37.08-Чистка и окраска тканей
	37.09-Manufacture of umbrellas, canes, fans, neckties, etc.	37.09-Производство зонтиков, тростей, вееров, галстуков и проч.
	37.10-Manufacture of wreaths, bouquets, artificial flowers, beads, imitation gems, etc.	37.10-Изготовление венков, букетов, искусственных цветов, бисера, поддельных камней и проч.
	37.11-Unspecified within group	37.11-Нераспределены внутри группы
38-Construction, repair and maintenance of residences and general construction work		38-Работы по устройству, ремонту и содержанию жилищ и строительные вообще
	38.01-Architects, builders, contractors, construction workers	38.01-Архитекторы, строители, подрядчики, рабочие строительных работ
	38.02-Bricklayers, diggers	38.02-Кладчики кирпича, землекопы
	38.03-Stonemasons, stone cutters	38.03-Каменщики, каменотесы
	38.04-Stove setters	38.04-Печники
	38.05-Carpenters	38.05-Плотники
	38.06-Caulkers of buildings	38.06-Конопатчики зданий
	38.07-Plasterers, stuccoists and mosaicists	38.07-Штукатуры, лепщики и мозаичисты
	38.08-Roofers	38.08-Кровельщики
	38.09-House painters	38.09-Маляры
	38.10-Glaziers	38.10-Стекольщики
	38.11-Technicians and workers who install gas, heating, plumbing, electrical and other appliances	38.11-Техники и рабочие по устройству газовых, нагревательных, водопроводных, электрических и т.п. приборов
	38.12-Floor polishers and chimney sweeps	38.12-Полотеры и трубочисты
	38.13-Upholstery	38.13-Обойщики и драпировщики
	38.14-Asphalt work	38.14-Асфальтовые работы
	38.15-Marble and gravestone cutters	38.15-Мраморщики и монументщики
	38.16-Marsh drainage, installation of artesian and other wells	38.16-Работы по осушению болот и устройство артезианских и др. колодцев
	38.17-Construction and repair of high and dirt roads, streets and bridges	38.17-Работы по постройке и ремонту шоссейных и грунтовых дорог, улиц и мостов на них
	38.18-Construction and repair of railways, lighthouses, docks, ports, breakwaters, etc.	38.18-Работы по постройке и ремонту железных дорог, маяков, доков, портов, молов и т.п.
	38.19-Unspecified within group	38.19-Нераспределены внутри группы
39-Manufacture of carriages and construction of wooden vessels		39-Производство экипажей и постройка деревянных судов
	39.01-Construction of wooden vessels	39.01-Постройка деревянных судов
	39.02-Manufacture of carriages	39.02-Производство экипажей
	39.03-Unspecified within group	39.03-Нераспределены внутри группы
40-Manufacture not included in above or undetermined		40-Участие в производствах, не вошедших в предыдущие группы или неопределенных
	40.01-Gas and electric lighting, water and electricity supply, etc.	40.01-Служащие в обществах газового и электрического освещения, водопроводных, доставления электрической энергии и др.

	40.02-Fabricators, white- and blue-collar workers at factories with unspecified nature of production	40.02-Фабриканты, служащие и рабочие на фабриках без определения рода производства
	40.03-Mechanics, technicians, machinists	40.03-Механики, техники и машинисты
	40.04-Stokers, greasers, etc.	40.04-Кочегары, смазчики и т.п.
	40.05-Unspecified within group	40.05-Нераспределены внутри группы
41-Water transport	41-Водные сообщения	
	41.01-Commercial shipping	41.01-Коммерческое судоходство
	41.02-Port and river administration, etc.	41.02-Портовая и речная администрация и т.д.
	41.03-Bargemen, towing vessels with horses	41.03-Бурлаки, тяга судов лошадьми
	41.04-Unspecified within group	41.04-Нераспределены внутри группы
42-Railroads	42-Железные дороги	
	42.01-Station masters, trainmasters, supervisors of locomotives and rolling stock, train inspectors, fare collectors and other such railway employees	42.01-Начальники станций, начальники служб движения, тяги, подвижных составов, поездные контролеры, кассиры и т.п. служащие на линиях ж.д.
	42.02-Conductors, engine drivers, roadmasters, clerks, etc.	42.02-Кондукторы, машинисты, дорожные мастера, конторщики и т.п.
	42.03-Workers and attendants on railways, weighers, station servants, etc.	42.03-Рабочие и служители на железных дорогах, весовщики, станционная прислуга и т.п.
	42.04-Unspecified within group	42.04-Нераспределены внутри группы
43-Carrier trade	43-Извозный промысел	
	43.01-Operation of state, rural and free post offices	43.01-Содержание казенных, земских и вольных почт
	43.02-Passenger carrier trade	43.02-Легковой извозный промысел
	43.03-Goods carrier trade	43.03-Ломовой извозный промысел
	43.04-Mining and metals carrier trade	43.04-Горнозаводский извозный промысел
	43.05-Floating and hauling of timber, firewood	43.05-Сплав и возка леса и дров
	43.06-Sanitation, sewage disposal	43.06-Ассенизация и вывозка нечистот
	43.07-Unspecified within group	43.07-Нераспределены внутри группы
44-Other overland transport and means of transport	44-Остальные сухопутные сообщения и средства передвижения	
	44.01-Maintenance of roads, streets, bridges	44.01-Содержание дорог, улиц и мостов
	44.02-Manual handling of loads (porters, stevedores, carriers, etc.)	44.02-Переноска тяжестей (дрягили, крючники, носильщики и т.п.)
	44.03-Horse-drawn railways and horse-drawn buses	44.03-Конножелезные дороги и омнибусы
	44.04-Unspecified within group	44.04-Нераспределены внутри группы
45-Mail, telegraph, and telephone	45-Почта, телеграф и телефон	
	45.01-Senior grade employees (postal service, telegraph, telephone)	45.01-Служащие высшего разряда (почта, телеграф и телефон)
	45.02-Telegraph and telephone operators, mail carriers	45.02-Телеграфисты, телеграфистки, телефонистки и почтальоны
	45.03-Attendants, guards and messengers (postal service, telegraph and telephone)	45.03-Служители, сторожа и рассыльные (почта, телеграф и телефон)
	45.04-Unspecified within group	45.04-Нераспределены внутри группы
46-Credit and public commercial institutions	46-Кредитные и общественные коммерческие учреждения	
	46.01-Credit institutions: banks, pawnbrokers, savings-and-loan associations, loan offices	46.01-Кредитные учреждения: банки, ломбарды, ссудосберегательные товарищества и кассы ссуд
	46.02-Insurance institutions	46.02-Страховые учреждения
	46.03-Unspecified within group	46.03-Нераспределены внутри группы

47-Commercial brokerage services		47-Торговое посредничество
	47.01-Large-scale commission activities; transport, classifieds and advertising offices; delivery services, information services, etc.; funeral services	47.01-Крупная комиссионная деятельность, конторы транспорт[ные], объявлений и реклам, доставл[яющие] занят[ия], справочн[ые] и т.п. Похоронные бюро
	47.02-Commission agents: factors, brokers, messengers and their artels; artel members, contractors, suppliers, etc.	47.02-Комиссионеры: факторы, маклаки, посыльные и артели их, артельщики, подрядчики-поставщики и т.п.
	47.03-Unspecified within group	47.03-Нераспределены внутри группы
48-Trade overall		48-Торговля вообще
	48.01-Large trading houses and firms in various specializations or without specifcation of specialization, their offices and agencies	48.01-Крупные торговые дома и фирмы разных специальностей или без определения специальности, их конторы и агентуры
	48.02-Trade in general, without specification of goods traded	48.02-Торговля вообще, без определения предметов торговли
	48.03-Small-scale, mixed trade in various goods and household items	48.03-Мелочная, смешанная торговля разными товарами и предметами домашнего обихода
	48.04-Unspecified within group	48.04-Нераспределены внутри группы
49-Trade in live livestock		49-Торговля живым скотом
	49.01-Trade in horses	49.01-Торговля лошадьми
	49.02-Trade in other livestock	49.02-Торговля прочим скотом
	49.03-Unspecified within group	49.03-Нераспределены внутри группы
50-Trade in grain products		50-Торговля зерновыми продуктами
	50.01-Trade in grain, flour, groats, etc.	50.01-Торговля зерном, мукой, крупой и т.п.
51-Trade in other agricultural products		51-Торговля остальными продуктами сельского хозяйства
	51.01-Purchase and resale of agricultural products overall	51.01-Скупка и перепродажа продуктов сельского хозяйства вообще
	51.02-Trade in linen, hemp and cotton	51.02-Торговля льном, пенькою, хлопком
	51.03-Trade in hay and straw	51.03-Торговля сеном и соломою
	51.04-Trade in vegetables, fruit, mushrooms	51.04-Торговля овощами, фруктами, грибами
	51.05-Trade in meat, game, poultry and fish	51.05-Торговля мясом, дичью, птицами и рыбой
	51.06-Trade in dairy products and eggs	51.06-Торговля молочными продуктами и яйцами
	51.07-Trade in baked bread	51.07-Торговля печеным хлебом
	51.08-Small-scale retailing of various foodstuffs and other consumables	51.08-Торговля мелочная, разными съестными припасами и др. предметами потребления
	51.09-Trade in colonial goods, groceries	51.09-Торговля колониальными и бакалейными товарами
	51.10-Trade in tobacco and tobacco products	51.10-Торговля табаком и табачными изделиями
	51.11-Trade in flowers	51.11-Торговля цветами
	51.12-Unspecified within group	51.12-Нераспределены внутри группы
52-Trade in construction materials and fuels		52-Торговля строительными материалами и топливом
	52.01-Trade in forest materials	52.01-Торговля лесным материалом
	52.02-Trade in other construction materials (stone, cement, etc.)	52.02-Торговля другими строительными материалами (камень, цемент и проч.)
	52.03-Trade in fuel (firewood, coal, peat, dry dung fuel)	52.03-Торговля топливом (дрова, уголь, торф, кизяк)

	52.04-Trade in cardboard, wallpaper, tar paper and paper, except writing paper	52.04-Торговля картоном, обоями, толем и бумагой, кроме писчей
	52.05-Trade in various home improvement goods (heaters, plumbing fixtures, etc.)	52.05-Торговля разными предметами устройства жилищ (нагревательные приборы, водопроводные принадлежности и т.п.)
	52.06-Unspecified within group	52.06-Нераспределены внутри группы
53-Trade in household articles		53-Торговля предметами домашнего обихода
	53.01-Trade in home improvement and household goods, hardware, etc.	53.01-Торговля предметами домашнего устройства и хозяйства, скобяным и т.п. товарами
	53.02-Trade in glassware and earthenware (porcelain, faience, etc.)	53.02-Торговля стеклянными и глиняными изделиями (фарфор, фаянс и проч.)
	53.03-Trade in furniture	53.03-Торговля мебелью
	53.04-Unspecified within group	53.04-Нераспределены внутри группы
54-Trade in metal goods, machines and weapons		54-Торговля разного рода металлическими товарами, машинами и оружием
	54.01-Trade in iron, steel, cast iron and other metals	54.01-Торговля железом, сталью, чугуном и др. металлами
	54.02-Trade in agricultural machinery	54.02-Торговля сельскохозяйственными машинами
	54.03-Trade in all types of machines (except agricultural) and bicycles	54.03-Торговля машинами всякого рода (кроме сельскохозяйственных) и велосипедами
	54.04-Trade in weapons and hunting accessories	54.04-Торговля оружием и принадлежностями охоты
	54.05-Unspecified within group	54.05-Нераспределены внутри группы
55-Trade in fabrics and garments		55-Торговля тканями и предметами одежды
	55.01-Trade in cloth and other manufactured textile fabrics and similar goods	55.01-Торговля сукнами и проч. тканями, мануфактурными, суровскими и т.п. товарами
	55.02-Trade in garments (underwear, ready-to-wear dresses, footwear, officer's articles, etc.)	55.02-Торговля предметами одежды (белье, готовое платье, обувь, офицерские вещи и проч.)
	55.03-Trade in various haberdashery goods	55.03-Торговля разными галантерейными товарами
	55.04-Unspecified within group	55.04-Нераспределены внутри группы
56-Trade in leather, fur, etc.		56-Торговля кожами, мехами и проч.
	56.01-Trade in leather, leather goods and saddlery	56.01-Торговля кожами, изделиями из них и шорным товаром
	56.02-Trade in furs	56.02-Торговля мехами
	56.03-Trade in down, feathers, bristles, wool and their products	56.03-Торговля пухом, пером, щетиной, шерстью и изделиями из них
	56.04-Unspecified within group	56.04-Нераспределены внутри группы
57-Trade in articles of luxury, science, art, culture, etc.		57-Торговля предметами роскоши, наук, искусств, культа и т.д.
	57.01-Operation of libraries and reading rooms, their workers and servants	57.01-Содержание библиотек и читален, служащие и прислуга при них
	57.02-Trade in books and notes; art objects	57.02-Торговля книгами и нотами; предметами искусств
	57.03-Trade in writing, drawing and drafting accessories, writing paper and toys	57.03-Торговля письменными, канцелярскими, рисовальными, чертежными принадлежностями, писчей бумагой и игрушками
	57.04-Trade in mathematical, physical and surgical instruments	57.04-Торговля приборами математическими, физическими и хирургическими
	57.05-Trade and rent of musical instruments	57.05-Торговля музыкальными инструментами, отдача их напрокат
	57.06-Trade in church goods	57.06-Торговля предметами церковными

	57.07-Trade in jewelry, gold and silver items	57.07-Торговля ювелирными, золотыми и серебряными изделиями
	57.08-Unspecified within group	57.08-Нераспределены внутри группы
58-Trade in other articles		58-Торговля остальными предметами
	58.01-Trade in chemical products (household, pharmaceutical, cosmetic, etc.)	58.01-Торговля химическими продуктами (москательными, аптекарскими, косметическими и проч.)
	58.02-Trade in illuminants (candles, kerosene, etc.)	58.02-Торговля осветительным материалом (свечи, керосин и т.п.)
	58.03-Trade in water for drinking and irrigation, watering troughs, stalls; trade in ice	58.03-Торговля водою для питья и орошения, водопойные колоды, будки, торговля льдом
	58.04-Trade in carriages, carts, etc.	58.04-Торговля экипажами, повозками и проч.
	58.05-Trade in vulcanized rubber goods	58.05-Торговля резиновыми изделиями
	58.06-Trade in various types of animals and birds	58.06-Торговля разного рода животными и птицами
	58.07-Unspecified within group	58.07-Нераспределены внутри группы
59-Peddling, street hawking		59-Торговля разносная и развозная
	59.01-Peddling, street hawking	59.01-Торговля разносная и развозная
60-Inns, hotels, rooming houses and clubs		60-Трактиры, гостиницы, меблированные комнаты, клубы
	60.01-Operation of hotels, rooming houses, dosshouses, inns	60.01-Содержание гостиниц, меблированных комнат, ночлежных домов, постоялых дворов
	60.02-Operation of restaurants, taverns, canteens, eating houses and tearooms	60.02-Содержание ресторанов, трактиров, буфетов, кухмистерских и чайных
	60.03-Operation and management of clubs and private theaters	60.03-Содержание и управление клубов и частных театров
	60.04-Operation of entertainment establishments, musical choirs and private orchestras	60.04-Содержание увеселительных заведений, музыкальных хоров, частных оркестров
	60.05-Pursuit of various types of sport as trade, operation of establishments for sport	60.05-Занятие разного рода спортом, как промыслом, и содержание заведений для спорта
	60.06-Keeping boarders and tenants	60.06-Содержание нахлебников и квартирантов
	60.07-Unspecified within group	60.07-Нераспределены внутри группы
61-Drinking establishments		61-Торговля питейная
	61.01-Wholesale wine trade, wine shops, wine cellars, etc.	61.01-Оптовая винная торговля, винные лавки, ренсковые погреба и т.д.
	61.02-Taphouses, pubs and other drinking establishments	61.02-Шинки, корчмы, портерные и другие заведения распивочной торговли
	61.03-Unspecified within group	61.03-Нераспределены внутри группы
62-Establishments providing bathing and personal hygiene services		62-Заведения, касающиеся чистоты и гигиены тела
	62.01-Saunas and bathhouses	62.01-Бани и купальни
	62.02-Barbers and hairdressers	62.02-Цирюльники и парикмахеры
	62.03-Laundries and ironing services	62.03-Прачечные и гладильни
	62.04-Unspecified within group	62.04-Нераспределены внутри группы
63-Persons of undetermined occupation		63-Лица неопределенных профессий
	63.01-Scribes, draftsmen, translators, petition writers, other freelance professionals	63.01-Переписчики, чертежники, переводчики, составители прошений и т.п. лица свободных профессий
	63.02-Wet nurse services: taking on nurslings from orphanages and private individuals for wet nursing and fostering	63.02-Питомнический промысел: принятие питомцев воспитательных домов на вскармливание и воспитание, а также и от частных лиц

	63.03-Travelling musicians, singers, acrobats, etc.	63.03-Бродячие музыканты, певцы, акробаты и т.п.
	63.04-Paupers, vagabonds, wayfarers, pilgrims, fortunetellers, etc.	63.04-Нищие, бродяги, странники, богомолки, гадалки и т.п.
	63.05-Unspecified within group	63.05-Нераспределены внутри группы
64-Prostitution		64-Проституция
	64.01-Operation of brothels	64.01-Содержание домов терпимости
	64.02-Prostitutes	64.02-Проститутки
	64.03-Unspecified within group	64.03-Нераспределены внутри группы
65-Unknown occupations		65-Неизвестные занятия
	65.01-Did not specify occupation or means of subsistence	65.01-Лица, не указавшие вовсе своих занятий или средств существования

Cross-section 2000

ENGLISH	RUSSIAN
Persons residing in residential social service institutions	Население, проживающее в стационарных учреждениях социального обслуживания
Persons residing in hospitals for persons suffering from chronic illnesses	Население, проживающее в больницах для лиц, страдающих хроническими заболеваниями
Persons residing in orphanages, infant orphanages and boarding schools for orphans and children without parental care	Население, проживающее в детских домах, домах ребенка, школах-интернатах для детей-сирот и детей, оставшихся без попечения родителей
Persons residing in other collective households	Население, проживающее в других коллективных домохозяйствах
Stipend recipients	Стипендиаты
Recipients of benefits (except unemployment benefits) and persons on social security	Лица, получающие пособие (кроме пособия по безработице), и лица, находящиеся на обеспечении государства
Subsisting on savings	Живущие за счёт сбережений
Subsisting on income from renting or leasing property	Живущие за счёт дохода от сдачи внаем или в аренда имущества
Unemployed	Безработные
Working, not for hire, without employment of hired workers (working in agriculture, hunting and forestry)	Работающие не по найму, без привлечения наёмных работников (работающие в сельском хозяйстве, охоте и лесном хозяйстве)
Working, not for hire, without employment of hired workers (working in agriculture, hunting and forestry)	Работающие не по найму, без привлечения наёмных работников (работающие в рыболовстве и рыбоводстве)
Did not specify employment status	Не указавшие положение в занятии
Working, not for hire - other	Работающие не по найму - иное
Working, not for hire, did not specify employment of hired workers	Работающие не по найму, не указавшие привлечение наемных работников
Working, not for hire, without employment of hired workers (except working in fishing industry and fish farming, agriculture, hunting and forestry)	Работающие не по найму, без привлечения наёмных работников (кроме работающих в рыболовстве, рыбоводстве, сельском хозяйстве, охоте и лесном хозяйстве)
Working, not for hire, with employment of hired workers	Работающие не по найму, с привлечением наемных работников
Working for hire	Работающие по найму

Appendix V – Authors

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Appendix VI – Licence Agreement

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    - Example – second citation:
      Global Collaboratory on the History of Labour Relations, Moll-Murata, China1800(June2011).